

A spectral gap for Metropolis-adjusted Langevin algorithm with a uniformly randomized step size

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Abstract

Let $\pi(dx) \propto e^{-U(x)} dx$ on $\mathbb{R}^d$, where U is continuously differentiable and m -strongly convex with a globally L -Lipschitz gradient, $0 < m \leq L < \infty$, and $\kappa = L/m$. Fixed-step Metropolis-adjusted Langevin algorithm (MALA) has known warm-start mixing-time upper bounds of order $\kappa\sqrt{d}$, up to logarithmic factors. A spectral-gap lower bound at the corresponding scale $(\kappa\sqrt{d})^{-1}$ would also control the asymptotic variance of sample averages uniformly over all square-integrable functions. However, for each fixed $\kappa > 1$, the best fixed-step spectral gap that can be guaranteed uniformly over this target class is at most of order $(\log d)/d$. We show that uniform randomization of the step size improves this worst-case guarantee. At each iteration, the algorithm draws $h \sim \text{Unif}(0, H)$ and performs one ordinary MALA transition. Choosing H of order $[L\sqrt{d(1 + \log d + \log \kappa)}]^{-1}$ yields a right spectral-gap lower bound of order

$$\frac{1}{\kappa\sqrt{d(1 + \log d + \log \kappa)}},$$

uniformly over the target class. For fixed $\kappa > 1$, this gives an upper bound of order $\sqrt{d \log d}$ on asymptotic variance relative to independent sampling, uniformly over all square-integrable functions. In contrast, every fixed step size has a worst-case variance ratio of at least order $d/\log d$. The gap also gives geometric convergence of the lazy kernel from every initial density in $L^2(\pi)$, central limit theorems, and nonstationary mean-square error bounds under an additional moment condition on the test function. The proof combines acceptance and flow estimates at different step-size scales using a Cheeger-type aggregation inequality. The scale used to control flow out of a set may depend on its stationary mass, and the aggregation inequality avoids the loss from first estimating the overall conductance and then applying the standard Cheeger inequality.

This work was developed with substantial assistance from ChatGPT, which suggested the uniformly randomized-step approach, developed the principal proof arguments, and generated the simulation and Lean 4 code. The human author checked and verified the mathematical content and takes full responsibility for the results.

1 Introduction

Markov chain Monte Carlo (MCMC) methods are sampling algorithms that can be used to estimate expectations under distributions from which independent sampling is difficult. The Metropolis-adjusted Langevin algorithm (MALA) uses the gradient of the log-density to construct proposals for this purpose. For a target probability distribution π on $\mathbb{R}^d$ with density proportional to e^{-U} , MALA proposes

$$Y = x - h\nabla U(x) + \sqrt{2h} Z, \quad Z \sim N_d(0, I_d),$$

and applies the Metropolis–Hastings correction. The step size h controls both the distance of a proposed move and its probability of acceptance. The geometric ergodicity and high-dimensional scaling of MALA were studied by [Roberts and Tweedie \[1996\]](#), [Roberts and Rosenthal \[1998\]](#). More recent work gives quantitative bounds on the number of iterations needed to approximate π . In particular, when U is m -strongly convex and has an L -Lipschitz gradient, [Wu et al. \[2022\]](#), strengthening [Chewi et al. \[2021\]](#), establish a warm-start mixing-time upper bound for lazy MALA with $\kappa\sqrt{d}$ dependence, up to logarithmic factors in d, κ , the warmness parameter M , and the desired total variation distance tolerance ϵ . Here, $\kappa = L/m$ is the condition number, and $M = \|\mathrm{d}\mu/\mathrm{d}\pi\|_{L^\infty(\pi)}$ where μ is the initial distribution. Their step size is of order $(L\sqrt{d})^{-1}$ up to logarithmic factors that again depend on d, k, M, ϵ . Their upper bound is accompanied by lower bounds that identify the $\kappa\sqrt{d}$ dependence, up to logarithmic factors, in the warm-start mixing problem considered there.

For Monte Carlo estimation, it is also crucial to control the dependence among successive samples; see, e.g., [Geyer \[1992\]](#), [Sokal \[1997\]](#), [Jones and Hobert \[2001\]](#), [Jones and Qin \[2022\]](#). For a stationary chain with kernel P , the asymptotic variance of a test function f is

$$\sigma_P^2(f) = \lim_{n \rightarrow \infty} n \operatorname{Var} \left(\frac{1}{n} \sum_{k=0}^{n-1} f(X_k) \right),$$

when the limit exists. For a reversible kernel with a positive right, or Poincaré, spectral gap, denoted by $\operatorname{Gap}_R(P)$, the standard spectral formula gives

$$\sup_{\substack{f \in L^2(\pi) \\ \operatorname{Var}_\pi(f) > 0}} \frac{\sigma_P^2(f)}{\operatorname{Var}_\pi(f)} = \frac{2}{\operatorname{Gap}_R(P)} - 1;$$

see [Kipnis and Varadhan \[1986\]](#), [Roberts and Rosenthal \[2008\]](#). The ratio on the left compares the long-run variance of an MCMC average with that of an average of independent draws from π . A lower bound on the spectral gap controls this ratio for every square-integrable test function.

This leads to the question of whether a MALA kernel admits a spectral-gap lower bound of order $(\kappa\sqrt{d})^{-1}$, up to logarithmic factors, uniformly over the target class described above. For a fixed step size $h > 0$, let P_h denote the MALA kernel. The desired bound would have the form

$$\operatorname{Gap}_R(P_h) \gtrsim \frac{1}{\kappa\sqrt{d}}, \tag{1}$$

where h may depend on d, m, L , and $\gtrsim$ here suppresses universal constants and logarithmic factors in d, κ . Besides a warm-start mixing-time upper bound, this would give a uniform bound of order $\kappa\sqrt{d}$, up to logarithmic factors, on the asymptotic variance relative to independent sampling. Prior to the current work, a fixed-step spectral-gap lower bound of order $(\kappa^2 d)^{-1}$ is available with a step size of order $(\kappa L d)^{-1}$ [[Liu and Tong, 2026](#)].

However, (1) cannot hold uniformly over the target class for a single fixed step size. Spectral-gap obstructions of this type were established in various regimes by [Chewi et al. \[2021\]](#), [Lee et al. \[2021\]](#), [Wu et al. \[2022\]](#). Using a modification of their hard-target constructions, [Proposition 2.3](#) gives an explicit worst-case spectral-gap upper bound for every choice of h . For each fixed $\kappa > 1$ and $h > 0$, the largest gap that can be guaranteed uniformly over the target class is at most of order $(\log d)/d$. Equivalently, no fixed choice of h gives a worst-case asymptotic variance ratio smaller than order $d/\log d$ as $d \rightarrow \infty$.

We study the uniformly randomized-step kernel

$$\overline{P}_H = \frac{1}{H} \int_0^H P_h \, \mathrm{d}h.$$

At each iteration, the algorithm draws $h \sim \text{Unif}(0, H)$ independently and performs one ordinary MALA transition. It uses one proposal per iteration, has essentially the same computational cost as fixed-step MALA, and is exactly π -reversible. Under strong convexity and a globally Lipschitz gradient, [Theorem 2.1](#) gives a positive spectral-gap lower bound for every $H > 0$. In particular, [Corollary 2.2](#) shows that choosing H of order

$$\frac{1}{L\sqrt{d(1 + \log d + \log \kappa)}}$$

gives a gap of at least order

$$\frac{1}{\kappa\sqrt{d(1 + \log d + \log \kappa)}}.$$

[Corollary 2.6](#) shows that the asymptotic variance $\sigma_{\bar{P}_H}^2(f)$ is at most a constant multiple of

$$\kappa\sqrt{d(1 + \log d + \log \kappa)} \text{Var}_\pi(f)$$

for every $f \in L^2(\pi)$. For fixed $\kappa > 1$, this is an upper bound of order $\sqrt{d \log d}$ on the variance ratio, whereas every fixed step size has a worst-case ratio of at least order $d/\log d$. [Corollary 2.7](#) states this comparison explicitly. The comparison is uniform over the target class and test functions; it does not assert that randomization reduces the variance for every individual target and function.

The spectral gap further supplies the hypotheses of standard results for Monte Carlo error. Under stationarity, every $f \in L^2(\pi)$ satisfies a central limit theorem [[Kipnis and Varadhan, 1986](#), [Roberts and Rosenthal, 2008](#)]. [Corollary 2.8](#) gives a finite-sample mean-square error bound for the lazy chain when the initial density belongs to $L^2(\pi)$ and $f \in L^4(\pi)$. This is derived using established error estimates for general Markov chains with spectral gaps [[Rudolf, 2012](#), [Paulin, 2015](#)].

A spectral gap also gives a warm-start mixing-time upper bound, but it provides this through a stronger geometric convergence result. For a lazy or positive semidefinite π -reversible kernel P and an initial distribution μ with $d\mu/d\pi \in L^2(\pi)$,

$$\begin{aligned} \|\mu P^n - \pi\|_{\text{TV}} &\leq \frac{1}{2} \left\| \frac{d(\mu P^n)}{d\pi} - 1 \right\|_{L^2(\pi)} \\ &\leq \frac{1}{2} \left\| \frac{d\mu}{d\pi} - 1 \right\|_{L^2(\pi)} [1 - \text{Gap}_R(P)]^n. \end{aligned} \tag{2}$$

See [Roberts and Rosenthal \[1997\]](#), [Kontoyiannis and Meyn \[2012\]](#), [Rudolf \[2012\]](#). Applied to $(I + \bar{P}_H)/2$, this yields [Corollary 2.5](#), which gives a warm-start mixing-time upper bound of order $\kappa\sqrt{d}$, up to logarithmic factors. Unlike warm-start mixing time bounds in [Chewi et al. \[2021\]](#), [Wu et al. \[2022\]](#), the step-size parameter H does not depend on the initial density or the desired accuracy. Thus, the spectral-gap lower bound certifies convergence all the way to π , without the need to tune H in accordance with a prescribed tolerance.

The proof of our spectral gap bound uses different step-size ranges to control flow out of different sets. This requires more than integrating known fixed-step spectral-gap lower bounds. Indeed, the usual inheritance inequality [[Grazzi et al., 2026](#)] gives

$$\text{Gap}_R(\bar{P}_H) \geq \frac{1}{H} \int_0^H \text{Gap}_R(P_h) dh,$$

but substituting available uniform fixed-step lower bounds does not give the desired rate. Each component gap $\text{Gap}_R(P_h)$ already takes an infimum over all test functions. Estimating these gaps

separately therefore loses the possibility of using different components to control different test functions.

We instead estimate the one-step flow out of a set $S \subset \mathbb{R}^d$, given by

$$\mathcal{J}_{\mathcal{K}_t}(S, S^c) = \int_S \mathcal{K}_t(x, S^c) \pi(\mathrm{d}x),$$

for the kernels $\mathcal{K}_t = (2/t) \int_{t/2}^t P_h \mathrm{d}h$, choosing the step scale $t \in (0, H]$ according to $\pi(S)$. For sets of moderate mass, relatively large steps have good acceptance outside an exceptional region small enough to be discarded. For smaller sets, we use stronger acceptance estimates at smaller scales. Isoperimetry converts these acceptance estimates into flow bounds. This way, for each set S satisfying $0 < \pi(S) < 1/2$, we control the flow out of S driven by MALA for a step size range $h \in (t/2, t) \subset (0, H)$.

Finally, a Cheeger-type aggregation lemma combines the flow bounds into a spectral-gap bound. This procedure avoids the loss incurred by first bounding the conductance of the mixture $\bar{P}_H$ and then applying the usual Cheeger inequality. The closest analytic predecessor of the aggregation lemma is the general Cheeger inequality of [Chen and Wang \[2000\]](#). We give a direct proof in [Lemma 3.5](#) and a formulation in [Theorem 3.6](#) that can be readily applied to our flow estimates.

As a byproduct of our analysis, we obtain an $L^p(\pi)$ bound on the rejection probability of fixed-step MALA. Let $r_h(x)$ denote the probability of rejecting a proposal from x with step size h . In [Proposition B.1](#) it is shown that, for some universal positive constants c, C ,

$$p \geq 1, 0 < h < \frac{c}{L\sqrt{p(d+p)}} \implies \|r_h\|_{L^p(\pi)} \leq CLh\sqrt{p(d+p)}.$$

This estimate does not require convexity of U . It complements existing MALA rejection bounds in the literature [[Chewi et al., 2021](#), [Wu et al., 2022](#), [Chen and Garmir, 2023](#)]. Its closest predecessor is the moment estimate in Lemma 17 of [Chewi et al. \[2021\]](#), which is stated under strong convexity of U .

An accompanying Lean 4 development [[Qin, 2026](#)] provides proof scripts for the main theorem and other principal conclusions, including [Theorems 2.1 and 3.6](#), [Proposition 2.3](#), and [Lemma 3.5](#). The scope of these scripts is described in [Section 5](#).

Relation to existing work

We now make some additional remarks regarding existing work relevant to our study.

Quantitative spectral-gap bounds for MALA exist in the literature, but are less common than mixing-time bounds. The classical work of [Roberts and Tweedie \[1996\]](#) gives qualitative criteria for geometric ergodicity. [Bou-Rabee and Hairer \[2013\]](#) show that, for drifts that are not globally Lipschitz, MALA may fail to have a spectral gap even when the underlying Langevin diffusion does. In the strongly convex, gradient-Lipschitz setting, [Lee et al. \[2021, Theorem 2\]](#) give a fixed-step gap upper bound of order $\log d/(\kappa d)$ under their restriction $\kappa = O(d^4)$; see also [Chewi et al. \[2021\]](#), [Wu et al. \[2022\]](#). As mentioned earlier, [Liu and Tong \[2026\]](#) obtain a spectral-gap lower bound of order $(\kappa^2 d)^{-1}$ with a step size of order $(\kappa L d)^{-1}$.

Mixing from initial distributions whose warmness grows exponentially with dimension has also been studied. For example, the Gaussian distribution $N_d(x_\star, L^{-1}I_d)$, where $x_\star$ is the minimizer of U , has warmness at most $\kappa^{d/2}$. From this initialization, [Chen et al. \[2020, Theorem 5\]](#) obtain an L^2 mixing-time bound of order $\max\{\kappa d, \kappa^{3/2}\sqrt{d}\} \log(d/\epsilon)$ for lazy MALA, with a step size independent of ϵ . [Lee et al. \[2020\]](#) obtain related guarantees using blocking conductance and a procedure for improving a constant-accuracy bound. Like in [Chewi et al. \[2021\]](#), [Wu et al. \[2022\]](#), the mixing-time

upper bound herein works best when the warmness grows sub-exponentially with d . Altschuler and Chewi [2024] provided a way to construct such warm starts using underdamped Langevin Monte Carlo before running MALA.

Randomized tuning parameters have previously been studied as a way to improve robustness. Grazi et al. [2026] study mixtures $\int P_h \nu(dh)$, establish inheritance of spectral gaps and weak Poincaré inequalities, and show that randomization can reduce sensitivity to step-size misspecification. Related benefits are known for Hamiltonian methods [Bou-Rabee and Sanz-Serna, 2017, Apers et al., 2024]. Our result concerns a dimension- and condition-number-explicit global gap for uniform-random MALA over the full strongly convex, gradient-Lipschitz target class.

Several established methods enter the acceptance analysis. Chewi et al. [2021] use a projection characterization of Metropolis adjustment and compare the Euler proposal with exact Langevin diffusion. Wu et al. [2022] compare one leapfrog step with continuous Hamiltonian dynamics; the relation between MALA and Hamiltonian methods is also discussed in Neal [2011], Chen and Gatmiry [2023]. Earlier rejection estimates and Wasserstein convergence bounds for perturbations of Gaussian measures appear in Eberle [2014] under higher-order regularity assumptions. Mangoubi and Vishnoi [2019] analyze Hamiltonian energy error under third- and fourth-order regularity conditions, including for nonconvex targets. Pathwise comparisons between Metropolized integrators and diffusions also appear in Bou-Rabee and Vanden-Eijnden [2010], Bou-Rabee and Hairer [2013].

Our appendix follows the diffusion-comparison approach. The stationary increment estimate follows from the Lyons–Zheng forward–backward decomposition [Lyons and Zheng, 1988, Section 1, especially equation (1.7)]; see also Fukushima et al. [2011, Theorem 5.7.1]. We give an elementary Langevin proof by subtracting the forward and reversed stochastic differential equations to cancel the drift. A Girsanov change of path law then compares the exact diffusion with the process whose drift is frozen at its initial state. Endpoint conditioning and the Metropolis–Hastings meet identity [Tierney, 1998, Billera and Diaconis, 2001] give an $L^p(\pi)$ rejection bound. The close-coupling and isoperimetric arguments use methods familiar from Metropolis analyses [Dwivedi et al., 2019, Wu et al., 2022, Andrieu et al., 2024, Liu and Tong, 2026]. The use of set mass to select a scale is related to average conductance, evolving sets, blocking conductance, and spectral profiles [Lovász and Kannan, 1999, Morris and Peres, 2005, Goel et al., 2006, Kannan et al., 2006, Montenegro, 2007].

A complementary line of work studies alternative gradient-based samplers, including randomized-midpoint discretizations of underdamped Langevin dynamics [Shen and Lee, 2019] and the path-space rejection methods of Chen et al. [2026]. Under their assumptions, the latter achieve expected gradient-query complexity with $\kappa^{2/3}d^{1/3}$ dependence, up to logarithmic factors, for sampling to a prescribed Rényi-divergence tolerance. These algorithms address the complexity of producing a sample at a specified accuracy. Our result concerns the convergence and Monte Carlo variance of a randomized-step MALA kernel, and is not a claim of optimal query complexity among all gradient-based samplers.

Organization

The rest of the article is organized as follows. Section 2 states the main bound, the fixed-step obstruction, and the consequences for mixing, asymptotic variance, and nonstationary Monte Carlo error, including the variance comparison in Corollary 2.7. Section 3 develops the ingredients of the proof, which is completed in Section 4. Section 5 describes the Lean development, and Section 6 gives numerical comparisons of fixed-step and randomized-step MALA. Technical proofs are collected in the Appendices.

Generative AI disclosure

This project began when the author asked ChatGPT 5.6 Sol (OpenAI) whether MALA admits a useful spectral-gap bound at the $d^{-1/2}$ step-size scale. ChatGPT suggested studying a uniformly randomized-step version of MALA—an algorithmic variant that had previously appeared in the literature—and developed the principal proof strategy and initial versions of the major mathematical arguments in this article. It also assisted in drafting and revising the manuscript and generated all code used for the numerical experiments and the Lean 4 formal verification. The human author checked the relevant literature, scrutinized and revised the AI-generated arguments, verified the mathematical claims and computations, inspected the simulation methodology and results, and confirmed that the Lean development was kernel-checked without proof placeholders or problem-specific axioms. The human author takes full responsibility for the content and correctness of the article.

2 Setup and Main Results

Assume throughout, except where explicitly stated otherwise, that d is a positive integer and that $U : \mathbb{R}^d \rightarrow \mathbb{R}$ is continuously differentiable, m -strongly convex, and has a globally L -Lipschitz gradient, where $0 < m \leq L < \infty$. Explicitly, for every $x, y \in \mathbb{R}^d$,

$$\begin{aligned} U(y) &\geq U(x) + \langle \nabla U(x), y - x \rangle + \frac{m}{2} \|y - x\|^2, \\ \|\nabla U(y) - \nabla U(x)\| &\leq L \|y - x\|. \end{aligned} \tag{3}$$

Strong convexity implies that e^{-U} is integrable. Let $\kappa := L/m \geq 1$, and define the target distribution

$$\pi(\mathrm{d}x) = \frac{e^{-U(x)}}{\int_{\mathbb{R}^d} e^{-U(u)} \mathrm{d}u} \mathrm{d}x.$$

For $h > 0$, let $Q_h(x, \cdot)$ be the Gaussian law

$$N_d(x - h\nabla U(x), 2hI_d).$$

The fixed-step MALA kernel is

$$P_h(x, \mathrm{d}y) = \alpha_h(x, y) Q_h(x, \mathrm{d}y) + \left[1 - \int \alpha_h(x, z) Q_h(x, \mathrm{d}z) \right] \delta_x(\mathrm{d}y),$$

where

$$\alpha_h(x, y) = 1 \wedge \frac{e^{-U(y)} q_h(y, x)}{e^{-U(x)} q_h(x, y)}$$

and q_h is the density of Q_h . The usual Metropolis–Hastings argument makes P_h π -reversible [Tierney, 1998]; consequently, so is the randomized-step MALA kernel

$$\overline{P}_H(x, \mathrm{d}y) = \frac{1}{H} \int_0^H P_h(x, \mathrm{d}y) \mathrm{d}h.$$

Thus H is fixed, while a new step size h is drawn independently at each iteration. The Metropolis–Hastings correction uses the proposal density for the step size drawn in that iteration.

For probability measures μ, ν on a measurable space $(\Omega, \mathcal{F})$, write

$$\mathrm{TV}(\mu, \nu) = \sup_{A \in \mathcal{F}} |\mu(A) - \nu(A)|.$$

For a probability measure ϖ , let $L^2(\varpi)$ be the space of measurable functions $f : \Omega \rightarrow \mathbb{R}$ such that $\int f^2 d\varpi < \infty$. For a ϖ -reversible kernel P and $f \in L^2(\varpi)$, define the Dirichlet form

$$\mathcal{E}_P(f, f) = \frac{1}{2} \int [f(x) - f(y)]^2 \varpi(dx) P(x, dy).$$

The right, or Poincaré, spectral gap is defined as

$$\text{Gap}_R(P) = \inf_{\substack{f \in L^2(\varpi) \\ \text{Var}_\varpi(f) > 0}} \frac{\mathcal{E}_P(f, f)}{\text{Var}_\varpi(f)}. \quad (4)$$

This quantity lies in $[0, 2]$. For a lazy or positive semidefinite kernel, it gives the geometric convergence bound in (2). For a general reversible kernel, a positive right gap also controls asymptotic variance; laziness is not needed for this conclusion.

The following theorem gives a spectral-gap lower bound for the randomized kernel. The constants are uniform over all potentials satisfying (3).

Theorem 2.1. *There is a universal constant $c_0 \in (0, \infty)$ such that the following holds. Define*

$$p_\star = 1 + \log d + \log \kappa.$$

Then, for every $H \in (0, \infty)$,

$$\text{Gap}_R(\bar{P}_H) \geq c_0 \frac{m}{H} \min \left\{ H, \frac{1}{L} \max \left\{ \frac{1}{\sqrt{p_\star(d + p_\star)}}, \frac{1}{d} \right\} \right\}^2. \quad (5)$$

The right spectral gap of the lazy kernel $(I + \bar{P}_H)/2$ is half of that of $\bar{P}_H$.

The bound is positive for every $H > 0$. As a function of H , it grows linearly until H reaches the second entry of the minimum in (5), and then decreases as H^{-1} . The next corollary gives the dimension dependence for two useful choices of H .

Corollary 2.2. *Under the notations of Theorem 2.1, let $c \in (0, \infty)$, and let $H = c/(L\sqrt{d})$. Then*

$$\text{Gap}_R(\bar{P}_H) \geq \frac{c_0}{\kappa\sqrt{d}} \min \left\{ c, \frac{1}{c} \max \left\{ \frac{d}{p_\star(d + p_\star)}, \frac{1}{d} \right\} \right\},$$

where $c_0 \in (0, \infty)$ is a universal constant. In particular, since $\min\{p_\star(d + p_\star)/d, d\} \leq 2p_\star$,

$$\text{Gap}_R(\bar{P}_H) \geq \frac{c_0}{\kappa\sqrt{d}} \min \left\{ c, \frac{1}{cp_\star} \right\},$$

after adjusting the universal constant c_0 . Choosing $c = c'/\sqrt{p_\star}$ for some $c' \in (0, \infty)$ gives

$$\text{Gap}_R(\bar{P}_H) \geq \frac{c_0}{\kappa\sqrt{d}p_\star} \min \left\{ c', \frac{1}{c'} \right\}.$$

Taking c' to be a fixed positive constant gives a gap lower bound of order $(\kappa\sqrt{d}p_\star)^{-1}$. With c fixed instead, the larger endpoint $H = c/(L\sqrt{d})$ gives a lower bound of order $(\kappa\sqrt{d}p_\star)^{-1}$. Thus the smaller choice of H incurs a square-root logarithmic loss, and the larger choice incurs a full logarithmic loss. Both choices are independent of the desired accuracy. The following result, proved in Section A, shows why a comparable uniform gap cannot be obtained by choosing a single fixed step size.

Proposition 2.3. Fix $\kappa_0 > 1$. There are a universal constant $c \in (0, \infty)$ and a constant $C \in (0, \infty)$ depending only on κ_0 such that the following holds. Let $d \geq 2$, $0 < m < L$, and $\kappa = L/m \geq \kappa_0$. Then

$$\sup_{h>0} \inf_{\substack{U \in C^\infty(\mathbb{R}^d) \\ mI_d \leq \nabla^2 U(x) \leq LI_d, x \in \mathbb{R}^d}} \text{Gap}_R(P_{U,h}) \leq C \max \left\{ \frac{\log(\kappa d)}{\kappa d}, e^{-cd} \right\}. \quad (6)$$

Here $P_{U,h}$ denotes the MALA kernel with step size h and target density proportional to e^{-U} .

Remark 2.4. In [Section G](#), it is shown that, when U satisfies (3), it holds that $\text{Gap}_R(P_{1/(2Ld)}) \geq c(\kappa d)^{-1}$ for some universal constant $c \in (0, \infty)$, so the upper bound in [Proposition 2.3](#) can be matched with a lower bound.

The order of the supremum and infimum in (6) is important: for each choice of h , there is a hard potential, which may depend on h . The result concerns selecting a step size uniformly over the target class. For each fixed $\kappa > 1$, it bounds the best such gap by a constant multiple of $(\log d)/d$ as $d \rightarrow \infty$, whereas [Corollary 2.2](#) gives the randomized kernel a uniform lower bound of order $(d \log d)^{-1/2}$. This separation is polynomial in dimension up to logarithmic factors. The hard-target construction builds on [Chewi et al. \[2021\]](#), [Lee et al. \[2021\]](#), [Wu et al. \[2022\]](#); our modification gives the explicit dependence in (6) for all $\kappa \geq \kappa_0 > 1$.

[Corollary 2.2](#) and (2) give a mixing-time bound. Let $\mu \ll \pi$ be a probability distribution on $\mathbb{R}^d$ with $d\mu/d\pi \in L^2(\pi)$. For a Markov kernel P with stationary distribution π and $\epsilon > 0$, define

$$n_{\text{mix}}(P, \mu, \epsilon) = \inf \{n \in \{0, 1, \dots\} : \|\mu P^n - \pi\|_{\text{TV}} \leq \epsilon\}, \quad \inf \emptyset = \infty.$$

Then the following holds.

Corollary 2.5. Under the notations of [Theorem 2.1](#), let $H = c/(L\sqrt{dp_\star})$, where $c \in (0, \infty)$, and put

$$\ell_{\mu, \epsilon} = \log \max \left\{ 1, \frac{1}{2\epsilon} \left\| \frac{d\mu}{d\pi} - 1 \right\|_{L^2(\pi)} \right\}.$$

Then

$$n_{\text{mix}} \left(\frac{I + \bar{P}_H}{2}, \mu, \epsilon \right) \leq \left\lceil \frac{2\ell_{\mu, \epsilon}}{\text{Gap}_R(\bar{P}_H)} \right\rceil \leq \left\lceil C\ell_{\mu, \epsilon} \kappa \sqrt{dp_\star} \right\rceil,$$

where $C \in (0, \infty)$ is $\max\{c, c^{-1}\}$ multiplied by a universal constant.

Proof. Since $0 < \text{Gap}_R(\bar{P}_H)/2 \leq 1$, the inequality $1 - x \leq e^{-x}$ for $x \in [0, 1]$, together with (2) and [Corollary 2.2](#) gives the desired mixing time bound. The logarithm defining $\ell_{\mu, \epsilon}$ is finite and nonnegative, including when $\mu = \pi$. If $\ell_{\mu, \epsilon} = 0$, the initial total-variation error is already at most ϵ , so the mixing time is zero. In the edge case $\text{Gap}_R(\bar{P}_H)/2 = 1$, (2) shows that the chain has law π after one step from μ ; the same bound remains valid. $\square$

The bound in [Corollary 2.5](#) holds for every initial density in $L^2(\pi)$, using the same H and the same geometric rate. A bounded density ratio is not required.

For comparison, [Wu et al. \[2022, Theorem 1\]](#) use a fixed step size of order

$$[L\sqrt{d} \log^2(\max\{\kappa, d, M/\epsilon, c\})]^{-1}$$

for an M -warm start (i.e., $d\mu/d\pi \leq M$), where $c > 0$ is universal. Their acceptance estimates control the chain outside an exceptional region whose allowed size depends on M and ϵ . Decreasing this size can require a smaller step. Thus that theorem gives a family of mixing guarantees with

tolerance-dependent tuning, whereas [Corollary 2.5](#) follows from a geometric L^2 bound for one kernel. This is a distinction between the guarantees, and does not mean that a fixed-step MALA chain must be retuned to continue converging.

A second consequence concerns Monte Carlo averages. For a stationary chain $(X_k)_{k \geq 0}$ with kernel P , write

$$\sigma_P^2(f) = \lim_{n \rightarrow \infty} n \operatorname{Var} \left(\frac{1}{n} \sum_{k=0}^{n-1} f(X_k) \right)$$

when this limit exists, allowing the value $+\infty$. The next corollary gives a finite bound and a central limit theorem. It is a consequence of the standard spectral formula for asymptotic variance [[Kipnis and Varadhan, 1986](#), [Paulin, 2015](#)]. Its role is to make the Monte Carlo implication of the spectral gap explicit.

Corollary 2.6. *Under the notation of [Theorem 2.1](#), let $H = c/(L\sqrt{d}p_\star)$, where $c \in (0, \infty)$. For every real-valued measurable function $f \in L^2(\pi)$, the stationary asymptotic variance exists and satisfies*

$$\sigma_{\bar{P}_H}^2(f) \leq \left(\frac{2}{\operatorname{Gap}_R(\bar{P}_H)} - 1 \right) \operatorname{Var}_\pi(f) \leq C\kappa\sqrt{d}p_\star \operatorname{Var}_\pi(f), \quad (7)$$

where $C \in (0, \infty)$ is $\max\{c, c^{-1}\}$ multiplied by a universal constant. Moreover, if $(X_k)_{k \geq 0}$ has kernel $\bar{P}_H$ and an arbitrary initial probability distribution μ on $\mathbb{R}^d$, then

$$\frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \left[f(X_k) - \int f d\pi \right] \xrightarrow{d} N(0, \sigma_{\bar{P}_H}^2(f)).$$

For the half-lazy kernel $(I + \bar{P}_H)/2$, the corresponding statements hold with its own stationary asymptotic variance:

$$\sigma_{(I+\bar{P}_H)/2}^2(f) \leq \left(\frac{4}{\operatorname{Gap}_R(\bar{P}_H)} - 1 \right) \operatorname{Var}_\pi(f) \leq 2C\kappa\sqrt{d}p_\star \operatorname{Var}_\pi(f).$$

Proof. Let $f_0 = f - \int f d\pi$. Regard $\bar{P}_H$ as a self-adjoint contraction on $L^2(\pi)$. On the mean-zero subspace $\{g \in L^2(\pi) : \int g d\pi = 0\}$, the definition of the spectral gap (4) gives $I - \bar{P}_H \geq \operatorname{Gap}_R(\bar{P}_H)I > 0$. Thus its inverse exists there and

$$0 \leq (I - \bar{P}_H)^{-1} \leq \operatorname{Gap}_R(\bar{P}_H)^{-1}I.$$

The spectral formula for stationary asymptotic variance [[Kipnis and Varadhan, 1986](#), [Paulin, 2015](#)] gives

$$\begin{aligned} \sigma_{\bar{P}_H}^2(f) &= 2 \left\langle f_0, (I - \bar{P}_H)^{-1} f_0 \right\rangle_{L^2(\pi)} - \|f_0\|_{L^2(\pi)}^2 \\ &\leq \left(\frac{2}{\operatorname{Gap}_R(\bar{P}_H)} - 1 \right) \operatorname{Var}_\pi(f). \end{aligned}$$

[Corollary 2.2](#) gives the final inequality in (7).

We next verify Harris ergodicity. Since $q_h(x, y) > 0$ and $\alpha_h(x, y) > 0$ for every $h > 0$ and $x, y \in \mathbb{R}^d$, $\bar{P}_H(x, A) > 0$ whenever $\pi(A) > 0$. Hence $\bar{P}_H$ is π -irreducible. It is also aperiodic: a period greater than one would require a set D with $\pi(D) > 0$ and $\bar{P}_H(x, D) = 0$ for π -almost every $x \in D$, contradicting the preceding inequality.

Let A be measurable with $\pi(A) = 1$. Since π has an everywhere positive density, every accepted proposal belongs to A almost surely. Write $r(x) = \bar{P}_H(x, \{x\}) < 1$ for the rejection probability. For $x \notin A$, the chain remains at x until its first accepted move, and therefore

$$\mathbb{P}_x(X_1 \notin A, \dots, X_n \notin A) = r(x)^n \longrightarrow 0.$$

For $x \in A$, we have $\bar{P}_H(x, A) = 1$. Thus every set of full π -measure is reached almost surely from every state. By [Roberts and Rosenthal \[2006, Theorem 6\(v\)\]](#), $\bar{P}_H$ is Harris recurrent. Its invariant probability distribution is π , so it is positive Harris recurrent and aperiodic.

Under stationarity, the finite variance bound and reversibility give the central limit theorem by [Roberts and Rosenthal \[2008, Theorem 7\]](#); see also [Kipnis and Varadhan \[1986\]](#). For a Harris ergodic chain, a central limit theorem under stationarity holds from every initial distribution with the same limiting variance; see [Jones \[2004, Remark 6\]](#). This proves the assertion for arbitrary μ .

Finally, $(I + \bar{P}_H)/2$ is reversible and has right spectral gap $\text{Gap}_R(\bar{P}_H)/2$. It has the same properties used above: its accepted moves have a positive density, and its holding probability is $[1 + r(x)]/2 < 1$ at every state. Applying the same arguments proves the half-lazy assertions. $\square$

For independent samples, the asymptotic variance is $\text{Var}_\pi(f)$. Thus [Corollary 2.6](#) bounds the variance inflation uniformly over the target class and all square-integrable observables. The same value of H is used for every observable.

The next corollary compares the worst-case asymptotic variance for P_h and $\bar{P}_H$.

Corollary 2.7. *Fix $\kappa_0 > 1$. There are universal constants $a, C > 0$ and a constant $c > 0$ depending only on κ_0 such that the following holds. Let $d \geq 2$, $0 < m < L$, and $\kappa = L/m \geq \kappa_0$, and write $p_\star = 1 + \log d + \log \kappa$.*

(i) *For $H = 1/(L\sqrt{dp_\star})$, every potential U satisfying (3), and every $f \in L^2(\pi)$,*

$$\sigma_{\bar{P}_H}^2(f) \leq C\kappa\sqrt{dp_\star} \text{Var}_\pi(f).$$

(ii) *For every $h > 0$, there are a smooth potential U satisfying (3) and a function $f \in L^2(\pi_U)$ such that*

$$\text{Var}_{\pi_U}(f) = 1, \quad \sigma_{P_{U,h}}^2(f) \geq c \min \left\{ \frac{\kappa d}{\log(\kappa d)}, e^{ad} \right\}.$$

Here π_U is the target associated with U . The asymptotic variance in (ii) is allowed to be infinite.

Proof. Assertion (i) follows directly from [Corollary 2.6](#).

For (ii), recall the spectral identity

$$\sup_{\substack{g \in L^2(\pi) \\ \text{Var}_\pi(g) > 0}} \frac{\sigma_P^2(g)}{\text{Var}_\pi(g)} = \frac{2}{\text{Gap}_R(P)} - 1 \tag{8}$$

for a reversible kernel P with a unique invariant distribution π , where the right side is interpreted as $+\infty$ when $\text{Gap}_R(P) = 0$; see [Kipnis and Varadhan \[1986\]](#), [Roberts and Rosenthal \[2008\]](#).

Fix $h > 0$. By [Proposition 2.3](#), there are a universal constant $a > 0$ and a constant $C_1 > 0$ depending only on κ_0 such that one can choose a smooth admissible potential U with

$$\text{Gap}_R(P_{U,h}) \leq 2C_1 \max \left\{ \frac{\log(\kappa d)}{\kappa d}, e^{-ad} \right\}.$$

The factor 2 allows for the possibility that the infimum over U in that proposition is not attained.

We also have $\text{Gap}_R(P_{U,h}) \leq 1$. Taking f in (4) to be the indicator of a measurable set A satisfying $0 < \pi_U(A) < 1$ gives

$$\text{Gap}_R(P_{U,h}) \leq \frac{\int_A P_{U,h}(x, A^c) \pi_U(dx)}{\pi_U(A)[1 - \pi_U(A)]} \leq \frac{1}{1 - \pi_U(A)}.$$

Since π_U has a density, such sets can have arbitrarily small positive probability. Letting $\pi_U(A)$ decrease to zero proves the claim.

Consequently, (8) yields

$$\begin{aligned} \sup_{\substack{g \in L^2(\pi_U) \\ \text{Var}_{\pi_U}(g) > 0}} \frac{\sigma_{P_{U,h}}^2(g)}{\text{Var}_{\pi_U}(g)} &= \frac{2}{\text{Gap}_R(P_{U,h})} - 1 \\ &\geq \frac{1}{\text{Gap}_R(P_{U,h})} \\ &\geq \frac{1}{2C_1} \min \left\{ \frac{\kappa d}{\log(\kappa d)}, e^{ad} \right\}. \end{aligned}$$

By the definition of the supremum, there is a function f with variance ratio at least

$$\frac{1}{4C_1} \min \left\{ \frac{\kappa d}{\log(\kappa d)}, e^{ad} \right\}.$$

Rescaling this function to have stationary variance one proves (ii), with $c = 1/(4C_1)$. $\square$

For each fixed $\kappa > 1$, the upper bound in (i) is of order $\sqrt{d \log d}$, whereas the lower bound in (ii) is of order $d/\log d$ as $d \rightarrow \infty$. The target and test function in (ii) may depend on h . Thus this is a comparison of uniform worst-case guarantees, rather than a comparison for every fixed target and test function.

The next corollary extends the variance guarantee to a bound on the finite-sample mean-square error for the lazy randomized-step MALA. We use the half-lazy kernel so that its right gap is also its absolute spectral gap.

Corollary 2.8. *Under the notation of Theorem 2.1, let $H = c/(L\sqrt{dp_\star})$, where $c > 0$. Let $(X_k)_{k \geq 0}$ have kernel $(I + \bar{P}_H)/2$ and initial distribution $\mu \ll \pi$, with $d\mu/d\pi \in L^2(\pi)$. For $f \in L^4(\pi)$, define*

$$f_0 = f - \int f d\pi, \quad \bar{f}_n = \frac{1}{n} \sum_{k=0}^{n-1} f(X_k).$$

Then, there is a universal constant $c_0 \in (0, \infty)$ such that, for every integer $n \geq 1$,

$$\begin{aligned} \mathbb{E}_\mu \left[\left(\bar{f}_n - \int f d\pi \right)^2 \right] &\leq \frac{4/\text{Gap}_R(\bar{P}_H) - 1}{n} \text{Var}_\pi(f) + \frac{512}{n^2 \text{Gap}_R(\bar{P}_H)^2} \left\| \frac{d\mu}{d\pi} - 1 \right\|_{L^2(\pi)} \|f_0\|_{L^4(\pi)}^2 \\ &\leq \frac{4 \max\{c, c^{-1}\} \kappa \sqrt{dp_\star}}{c_0 n} \text{Var}_\pi(f) + \\ &\quad \frac{512 \max\{c^2, c^{-2}\} \kappa^2 dp_\star}{c_0^2 n^2} \left\| \frac{d\mu}{d\pi} - 1 \right\|_{L^2(\pi)} \|f_0\|_{L^4(\pi)}^2 \end{aligned} \tag{9}$$

Proof. The kernel $(I + \bar{P}_H)/2$ is reversible and positive semidefinite, with operator norm

$$1 - \frac{\text{Gap}_R(\bar{P}_H)}{2} < 1$$

on the mean-zero subspace of $L^2(\pi)$. Apply [Rudolf \[2012, Theorem 3.41\]](#) with $p = 4$, no burn-in, and the centered function f_0 . This gives

$$\begin{aligned} \mathbb{E}_\mu \left[\left(\bar{f}_n - \int f \, d\pi \right)^2 \right] &\leq \mathbb{E}_\pi \left[\left(\bar{f}_n - \int f \, d\pi \right)^2 \right] \\ &\quad + \frac{512}{n^2 \text{Gap}_R(\bar{P}_H)^2} \left\| \frac{d\mu}{d\pi} - 1 \right\|_{L^2(\pi)} \|f_0\|_{L^4(\pi)}^2. \end{aligned} \tag{10}$$

For the stationary chain,

$$0 \leq \left\langle f_0, \left(\frac{I + \bar{P}_H}{2} \right)^k f_0 \right\rangle_{L^2(\pi)} \leq \left(1 - \frac{\text{Gap}_R(\bar{P}_H)}{2} \right)^k \text{Var}_\pi(f).$$

Consequently,

$$\begin{aligned} \mathbb{E}_\pi \left[\left(\bar{f}_n - \int f \, d\pi \right)^2 \right] &= \frac{1}{n^2} \left[n \text{Var}_\pi(f) + 2 \sum_{k=1}^{n-1} (n-k) \left\langle f_0, \left(\frac{I + \bar{P}_H}{2} \right)^k f_0 \right\rangle_{L^2(\pi)} \right] \\ &\leq \frac{\text{Var}_\pi(f)}{n} \left[1 + 2 \sum_{k=1}^{\infty} \left(1 - \frac{\text{Gap}_R(\bar{P}_H)}{2} \right)^k \right] \\ &= \frac{4/\text{Gap}_R(\bar{P}_H) - 1}{n} \text{Var}_\pi(f). \end{aligned}$$

This along with (10) proves the first inequality in (9). The final inequality follows from [Corollary 2.2](#). $\square$

3 Ingredients of the Proof of [Theorem 2.1](#)

We now describe the main ingredients of the proof of [Theorem 2.1](#).

For $t > 0$, define the MALA kernel averaged over the upper dyadic half of $(0, t)$ by

$$\mathcal{K}_t = \frac{2}{t} \int_{t/2}^t P_h \, dh.$$

The uniform mixture $\bar{P}_H$ contains this component with weight $t/(2H)$ whenever $t \leq H$. The following elementary relation holds.

Lemma 3.1. *Let $0 < t_J < \dots < t_0 \leq H$, and suppose that the intervals $(t_j/2, t_j)$, $j \in \{0, \dots, J\}$, are pairwise disjoint. Then, for $f \in L^2(\pi)$,*

$$\mathcal{E}_{\bar{P}_H}(f, f) \geq \sum_{j=0}^J \frac{t_j}{2H} \mathcal{E}_{\mathcal{K}_{t_j}}(f, f).$$

Proof. By nonnegativity and Tonelli's theorem,

$$\mathcal{E}_{\overline{P}_H}(f, f) = \frac{1}{H} \int_0^H \mathcal{E}_{P_h}(f, f) dh \geq \sum_{j=0}^J \frac{1}{H} \int_{t_j/2}^{t_j} \mathcal{E}_{P_h}(f, f) dh.$$

The definition of $\mathcal{K}_{t_j}$ gives the claim. $\square$

We will analyze $\mathcal{K}_t$ at different scale parameters t , and aggregate results along an appropriate sequence of scale levels in a way that is more sophisticated than [Lemma 3.1](#).

To be more specific, in [Proposition B.1](#), we bound the fixed-step MALA rejection rate, and use this bound to establish a close-coupling, or one-step overlap, result for $\mathcal{K}_t$ in [Proposition 3.2](#). The curvature assumption on U yields the isoperimetric inequality in [Proposition 3.3](#). We then combine [Propositions 3.2](#) and [3.3](#) in [Proposition 3.4](#) to bound the one-step flow for $\mathcal{K}_t$ out of a given set S . Importantly, for each set S , the flow out of S is controlled for one particular $\mathcal{K}_t$ where the scale parameter t depends on $\pi(S)$. The general component-aggregation bound in [Theorem 3.6](#) combines the one-step flows at various scales, yielding [Theorem 2.1](#).

3.1 Close coupling

The next proposition, proved in [Section B](#), shows that $\mathcal{K}_t(x, \cdot)$ and $\mathcal{K}_t(y, \cdot)$ are close in total variation whenever $\|x - y\| \leq \sqrt{t}/16$. For the larger admissible values of t , this conclusion is guaranteed only when x and y lie outside a small exceptional set.

Proposition 3.2. *There are universal constants $c_r, C_r > 0$ such that, for every real $p \geq 1$ and every t satisfying*

$$0 < t \leq \frac{c_r}{L\sqrt{p(d+p)}}, \quad (11)$$

there is a measurable set $G_{t,p} \subseteq \mathbb{R}^d$ satisfying

$$\pi(G_{t,p}^c) \leq \left[C_r L t \sqrt{p(d+p)} \right]^p, \quad (12)$$

and

$$x, y \in G_{t,p}, \quad \|x - y\| \leq \frac{\sqrt{t}}{16} \implies \text{TV}(\mathcal{K}_t(x, \cdot), \mathcal{K}_t(y, \cdot)) \leq \frac{3}{4}.$$

Moreover, if $t' \in (0, 1/(2Ld)]$, then

$$x, y \in \mathbb{R}^d, \quad \|x - y\| \leq \frac{\sqrt{t'}}{16} \implies \text{TV}(\mathcal{K}_{t'}(x, \cdot), \mathcal{K}_{t'}(y, \cdot)) \leq \frac{3}{4}.$$

3.2 Isoperimetric inequality

A strongly log-concave distribution satisfies the Gaussian enlargement inequality below; see [Bakry and Ledoux \[1996, Corollary 2.2 and equations \(2.10\)–\(2.12\)\]](#). It remains valid under the first-order assumptions [\(3\)](#). To see this without assuming second differentiability, note that $U(x) - m\|x\|^2/2$ is convex. Caffarelli's contraction theorem [[Caffarelli, 2000](#)], in the nonsmooth formulation of [Kim and Milman \[2012, Theorem 1.1, Lemma 3.3, and the conclusion of Section 3\]](#), therefore gives a 1-Lipschitz map T sending $N_d(0, m^{-1}I_d)$ to π . For each Borel set A and $r > 0$, the inclusion $(T^{-1}(A))^r \subseteq T^{-1}(A^r)$ transfers the Gaussian isoperimetric inequality to π . Consequently,

$$\pi(A^r) \geq \Phi(\Phi^{-1}(\pi(A)) + \sqrt{m}r), \quad (13)$$

where, for $r > 0$, $A^r = \{x : \text{dist}(x, A) < r\}$, and Φ is the standard Gaussian distribution function. The cases $\pi(A) = 0, 1$ are understood by continuity. The same bound holds for sets measurable in the completion of π , by choosing a Borel subset of the same measure. The following three-set form of the isoperimetric inequality is proved in [Section C](#).

Proposition 3.3. *Let A, B, C be a measurable partition of $\mathbb{R}^d$, let $r \geq 0$, and suppose $\text{dist}(A, B) \geq r$. If $0 < q = \min\{\pi(A), \pi(B)\} \leq 1/2$, then*

$$\pi(C) \geq \frac{q}{4} \min \left\{ 1, r \sqrt{m \log(1/q)} \right\}. \quad (14)$$

3.3 One-step flow

For a π -reversible Markov kernel K , define the one-step flow

$$\mathcal{J}_K(S, S^c) = \int_S K(x, S^c) \pi(dx).$$

Combining [Proposition 3.2](#) and [Proposition 3.3](#), we establish the following result concerning $\mathcal{K}_t$ in [Section D](#).

Proposition 3.4. *One can find universal constants $b_0 \in (0, 1/2]$ and $A_0 \in [2, \infty)$ such that the following holds. Let $p \geq \tilde{p}_\star := A_0 p_\star$, where $p_\star$ is defined in [Theorem 2.1](#). Let $\theta \in (0, 1]$, and*

$$t = \frac{\theta b_0}{L \sqrt{p(d+p)}}.$$

Then, for every measurable set $S \subseteq \mathbb{R}^d$ such that $e^{-p/2} \leq \pi(S) \leq 1/2$,

$$\mathcal{J}_{\mathcal{K}_t}(S, S^c) \geq 2^{-12} \pi(S) \sqrt{mt \log[1/\pi(S)]},$$

where $mt \log[1/\pi(S)] \leq 1$.

Moreover, for $t' \in (0, 1/(2Ld)]$ and every measurable set $S \subseteq \mathbb{R}^d$ such that $0 < \pi(S) \leq 1/2$,

$$\mathcal{J}_{\mathcal{K}_{t'}}(S, S^c) \geq 2^{-12} \pi(S) \min \left\{ 1, \sqrt{mt' \log[1/\pi(S)]} \right\}.$$

Using [Proposition 3.4](#), for a set $S \subset \mathbb{R}^d$ satisfying $0 < \pi(S) < 1/2$, we may bound the average flow $\mathcal{J}_{\mathcal{K}_t}(S, S^c)/\pi(S)$ for appropriate values of t . The smaller $\pi(S)$ is, the smaller t for which the average flow can be estimated via [Proposition 3.4](#).

3.4 Aggregation of one-step flows

In [Section 4](#), we choose disjoint intervals of the form $[t_j/2, t_j]$ contained in $[0, H]$, where $(t_0, \dots, t_J)$ is a decreasing sequence of positive numbers. For a given measurable set S such that $0 < \pi(S) \leq 1/2$, the average flow $\mathcal{J}_{\mathcal{K}_{t_j}}(S, S^c)/\pi(S)$ may be large for some values of j but small for others. We therefore assign to each set S an appropriate index $j(S)$. Then, we establish a lower bound $\phi_{j(S)}$ on $\mathcal{J}_{\mathcal{K}_{t_{j(S)}}}(S, S^c)/\pi(S)$ using [Proposition 3.4](#). The purpose of this subsection is to provide the tools for combining these set-dependent bounds into a lower bound on $\text{Gap}_R(\bar{P}_H)$.

Cheeger's inequality [[Lawler and Sokal, 1988](#)] implies that the spectral gap of a reversible Markov kernel can be estimated by considering the kernel's average flow out of measurable sets. It may be tempting to use [Lemma 3.1](#) with $f = \mathbf{1}_S$, where $0 < \pi(S) \leq 1/2$, to obtain

$$\frac{\mathcal{J}_{\bar{P}_H}(S, S^c)}{\pi(S)} \geq \frac{1}{\pi(S)} \sum_{j=0}^J \frac{t_j}{2H} \mathcal{J}_{\mathcal{K}_{t_j}}(S, S^c) \geq \frac{t_{j(S)}}{2H} \frac{\mathcal{J}_{\mathcal{K}_{t_{j(S)}}}(S, S^c)}{\pi(S)} \geq \frac{\min_j t_j \phi_j}{2H}.$$

This gives a lower bound on the conductance of $\bar{P}_H$. It turns out that this direct estimate followed by the usual Cheeger inequality loses too much and does not yield the spectral lower bound of order $(\kappa\sqrt{d})^{-1}$ that we desire. We carry out the more careful approach below.

The following lemma carries out this aggregation for a generic mixture of reversible Markov kernels. It is possible to derive the lemma from the general Cheeger inequality of [Chen and Wang \[2000\]](#). We give a direct proof in [Section E](#).

Lemma 3.5. *Let $P, K_1, \dots, K_N$ be Markov kernels that are reversible with respect to a distribution ϖ . Suppose that there exist constants $\gamma_1, \dots, \gamma_N > 0$ such that*

$$\mathcal{E}_P(f, f) \geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(f, f) \quad \text{for every } f \in L^2(\varpi).$$

Suppose further that there exist constants $\beta_1, \dots, \beta_N \geq 0$ satisfying

$$\sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} > 0$$

such that, for every measurable set S satisfying $0 < \varpi(S) \leq 1/2$,

$$\sum_{j=1}^N \beta_j \mathcal{J}_{K_j}(S, S^c) \geq \varpi(S),$$

where

$$\mathcal{J}_{K_j}(S, S^c) = \int_S K_j(x, S^c) \varpi(dx).$$

Then

$$\text{Gap}_R(P) \geq \left(2 \sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} \right)^{-1}.$$

The above lemma gives the following convenient hard-assignment formulation, which we use to combine the flow bounds in [Proposition 3.4](#). Roughly speaking, [Theorem 3.6](#) allows the kernel that supplies the useful one-step flow bound to depend on the set under consideration, while keeping track of how often each kernel is selected in the mixture.

Theorem 3.6. *Let $P, K_1, \dots, K_N$ be Markov kernels that are reversible with respect to a distribution ϖ . Suppose*

$$\mathcal{E}_P(f, f) \geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(f, f), \quad \gamma_j > 0,$$

for every $f \in L^2(\varpi)$.

Suppose further that there exist constants $\phi_1, \dots, \phi_N > 0$ such that every measurable set S with $0 < \varpi(S) \leq 1/2$ can be assigned an index $j(S) \in \{1, \dots, N\}$ satisfying

$$\mathcal{J}_{K_{j(S)}}(S, S^c) \geq \phi_{j(S)} \varpi(S).$$

Then

$$\text{Gap}_R(P) \geq \left(2 \sum_{j=1}^N \frac{1}{\gamma_j \phi_j^2} \right)^{-1}.$$

Proof. Apply [Lemma 3.5](#) with

$$\beta_j = \frac{1}{\phi_j}, \quad j = 1, \dots, N.$$

For every measurable S with $0 < \varpi(S) \leq 1/2$,

$$\begin{aligned} \sum_{j=1}^N \beta_j \mathcal{J}_{K_j}(S, S^c) &\geq \frac{\mathcal{J}_{K_{j(S)}}(S, S^c)}{\phi_{j(S)}} \\ &\geq \varpi(S). \end{aligned}$$

The conclusion follows directly from [Lemma 3.5](#). □

4 Proof of the Main Theorem

Proof of Theorem 2.1. Let $b_0 \leq 1/2$ and $A_0 \geq 2$ be given in [Proposition 3.4](#), and put $\tilde{p}_\star = A_0 p_\star$, where $p_\star$ is defined in [Theorem 2.1](#). We divide the proof into three parts.

Part 1: The geometric moment ladder.

Suppose for now that $\tilde{p}_\star < d$. Let

$$p_j = 4^j \tilde{p}_\star, \quad j = 0, 1, \dots, J, \tag{15}$$

where J is the smallest integer for which $p_J \geq d$. Thus

$$d \leq p_J < 4d. \tag{16}$$

Define the nominal scales

$$\tau_j = \frac{b_0}{L\sqrt{p_j(d + p_j)}},$$

and put

$$\theta = \min \left\{ 1, \frac{H}{\tau_0} \right\}, \quad t_j = \theta \tau_j. \tag{17}$$

Then $t_0 \leq H$, and

$$\frac{t_{j+1}}{t_j} = \frac{1}{2} \sqrt{\frac{d + p_j}{d + 4p_j}} < \frac{1}{2}.$$

Consequently, the Borel sets $[t_j/2, t_j]$ are pairwise disjoint, have Lebesgue measure $t_j/2$, and lie within $[0, H]$. Applying [Lemma 3.1](#) to this finite family gives

$$\mathcal{E}_{\overline{P}_H}(f, f) \geq \sum_{j=0}^J \frac{t_j}{2H} \mathcal{E}_{\mathcal{K}_{t_j}}(f, f). \tag{18}$$

This prompts us to apply [Theorem 3.6](#).

Let S be measurable with $0 < \pi(S) \leq 1/2$. We assign S to a component $\mathcal{K}_{t_j}$. That is, we identify the index $j(S)$ in [Theorem 3.6](#). Since $A_0 \geq 2$ and $p_\star \geq 1$,

$$\frac{p_0}{2} = \frac{\tilde{p}_\star}{2} \geq 1 > \log 2.$$

The intervals $[\log 2, p_0/2]$, $(p_{j-1}/2, p_j/2]$ for $1 \leq j \leq J$, and $(p_J/2, \infty)$ are disjoint and cover $[\log 2, \infty)$. Thus the following assignment is exhaustive and unambiguous.

- If $\log 2 \leq \log[1/\pi(S)] \leq p_0/2$, assign S to $\mathcal{K}_{t_0}$. By the first assertion of [Proposition 3.4](#),

$$\mathcal{J}_{\mathcal{K}_{t_0}}(S, S^c) \geq \phi_0 \pi(S), \quad \phi_0 = 2^{-12} \sqrt{mt_0 \log 2}.$$

- For $1 \leq j \leq J$, if $p_{j-1}/2 < \log[1/\pi(S)] \leq p_j/2$, assign S to $\mathcal{K}_{t_j}$. By the first assertion of [Proposition 3.4](#),

$$\mathcal{J}_{\mathcal{K}_{t_j}}(S, S^c) \geq \phi_j \pi(S), \quad \phi_j = 2^{-12} \sqrt{\frac{mt_j p_{j-1}}{2}} = 2^{-12} \sqrt{\frac{mt_j p_j}{8}}.$$

- If $\log[1/\pi(S)] > p_J/2$, assign S to the last component $\mathcal{K}_{t_J}$. By [\(16\)](#) and the assumption $b_0 \leq 1/2$,

$$t_J \leq \frac{b_0}{L\sqrt{p_J(d+p_J)}} \leq \frac{1}{2Ld}.$$

By the second assertion of [Proposition 3.4](#),

$$\mathcal{J}_{\mathcal{K}_{t_J}}(S, S^c) \geq \phi'_J \pi(S), \quad \phi'_J = 2^{-12} \min \left\{ 1, \sqrt{\frac{mt_J p_J}{2}} \right\}.$$

Moreover,

$$\frac{mt_J p_J}{2} \leq \frac{b_0}{2\kappa} \sqrt{\frac{p_J}{d+p_J}} \leq \frac{b_0}{2} < 1.$$

Weakening the flow bound by a constant factor, we obtain

$$\mathcal{J}_{\mathcal{K}_{t_J}}(S, S^c) \geq \phi_J \pi(S), \quad \phi_J = 2^{-12} \sqrt{\frac{mt_J p_J}{8}}.$$

We may therefore apply [Theorem 3.6](#) to [\(18\)](#), with

$$\gamma_j = \frac{t_j}{2H}, \quad \phi_0^2 = 2^{-24} mt_0 \log 2, \quad \phi_j^2 = 2^{-27} mt_j p_j \quad (j \geq 1).$$

The summation estimate in [Lemma F.1](#) yields

$$\sum_{j=0}^J \frac{1}{\gamma_j \phi_j^2} \leq C \frac{HL^2 \tilde{p}_*(d + \tilde{p}_*)}{m\theta^2 b_0^2},$$

where $C \in (0, \infty)$ is universal. Hence,

$$\text{Gap}_R(\bar{P}_H) \geq \frac{m\theta^2 b_0^2}{2CHL^2 \tilde{p}_*(d + \tilde{p}_*)} = \frac{m}{2CH} \min \left\{ H, \frac{b_0}{L\sqrt{\tilde{p}_*(d + \tilde{p}_*)}} \right\}^2. \quad (19)$$

Part 2: A conservative bound

The bound [\(19\)](#) is almost what we need, but it has been established only for $\tilde{p}_* < d$. To obtain the exact bound in [Theorem 2.1](#), we derive a lower bound on $\text{Gap}_R(\bar{P}_H)$ using only conservative step sizes.

Define the safe step size

$$t_s = \min \left\{ H, \frac{b_0}{Ld} \right\}.$$

Because $b_0 \leq 1/2$, $t_s \leq 1/(2Ld)$. For every measurable S with $0 < \pi(S) \leq 1/2$, apply the second assertion of [Proposition 3.4](#) to obtain

$$\mathcal{J}_{\mathcal{K}_{t_s}}(S, S^c) \geq 2^{-12} \pi(S) \min \left\{ 1, \sqrt{mt_s \log[1/\pi(S)]} \right\} \geq 2^{-12} \pi(S) \min \left\{ 1, \sqrt{mt_s \log 2} \right\}. \quad (20)$$

Since

$$mt_s \log 2 \leq \frac{b_0 \log 2}{\kappa d} < 1,$$

the bound (20) becomes

$$\mathcal{J}_{\mathcal{K}_{t_s}}(S, S^c) \geq 2^{-12} \pi(S) \sqrt{mt_s \log 2}.$$

By [Lemma 3.1](#),

$$\mathcal{E}_{\overline{P}_H}(f, f) \geq \frac{t_s}{2H} \mathcal{E}_{\mathcal{K}_{t_s}}(f, f).$$

Apply [Theorem 3.6](#) with $N = 1$, $K_1 = \mathcal{K}_{t_s}$, $\gamma_1 = t_s/(2H)$, and $\phi_1 = 2^{-12} \sqrt{mt_s \log 2}$ to obtain

$$\text{Gap}_R(\overline{P}_H) \geq 2^{-26} (\log 2) \frac{mt_s^2}{H} = 2^{-26} (\log 2) \frac{m}{H} \min \left\{ H, \frac{b_0}{Ld} \right\}^2. \quad (21)$$

Part 3: Completion of the proof

If $\tilde{p}_\star \geq d$, then

$$\frac{1}{\sqrt{\tilde{p}_\star(d + \tilde{p}_\star)}} \leq \frac{1}{d}.$$

If $\tilde{p}_\star < d$, combine (19) and (21), using

$$\max\{\min(H, a)^2, \min(H, b)^2\} = \min\{H, \max(a, b)\}^2 \quad (a, b, H > 0).$$

In either case, with the universal constant $c_1 = \min\{(2C)^{-1}, 2^{-26} \log 2\} > 0$, we obtain

$$\text{Gap}_R(\overline{P}_H) \geq c_1 \frac{m}{H} \min \left\{ H, \frac{b_0}{L} \max \left\{ \frac{1}{\sqrt{\tilde{p}_\star(d + \tilde{p}_\star)}}, \frac{1}{d} \right\} \right\}^2.$$

Since $A_0 \geq 1$ and $\tilde{p}_\star = A_0 p_\star$,

$$\tilde{p}_\star(d + \tilde{p}_\star) = A_0 p_\star(d + A_0 p_\star) \leq A_0^2 p_\star(d + p_\star),$$

and, because $b_0 \leq 1$, it follows that

$$\text{Gap}_R(\overline{P}_H) \geq c_1 \left(\frac{b_0}{A_0} \right)^2 \frac{m}{H} \min \left\{ H, \frac{1}{L} \max \left\{ \frac{1}{\sqrt{p_\star(d + p_\star)}}, \frac{1}{d} \right\} \right\}^2.$$

Choosing $c_0 = (c_1/2)(b_0/A_0)^2$ proves (5).

The identity kernel has zero Dirichlet form, so

$$\text{Gap}_R((I + \overline{P}_H)/2) = \frac{1}{2} \text{Gap}_R(\overline{P}_H).$$

□

5 Lean verification

The accompanying development in Lean 4 [de Moura and Ullrich, 2021], stored in the GitHub repository [Qin, 2026], provides proof scripts for the principal results of this article. Its first-order input consists of a continuously differentiable potential, the strong-convexity inequality in (3), and a Lipschitz bound on the gradient of that potential.

The supplied scripts connect this input to Theorem 2.1, and to Corollaries 2.2 and 2.5. The smooth hard-target construction and the fixed-step minimax statement in Propositions A.1 and 2.3 are retained. The proof of Theorem 2.1 uses the mixture comparison in Lemma 3.1, the overlap and separation bounds in Propositions 3.2 and 3.3, both flow bounds in Proposition 3.4, and the aggregation results in Lemma 3.5 and Theorem 3.6.

The stationary-rejection and overlap conclusions (Propositions B.1 and 3.2) are obtained through an alternative discrete-time proof, using finite-dimensional Gaussian likelihood estimates, an Euler/random-walk-Metropolis comparison, and weak-limit stability. The continuous-time proofs of Lemmas B.2 to B.5 are not transcribed into Lean. In particular, the additional nonconvex scope noted at the beginning of Appendix B is not part of the formalization: the Lean stationary-rejection theorem retains the standing strong-convexity assumption.

The isoperimetric bound is derived internally from the Gaussian Ornstein–Uhlenbeck argument and a finite-Euler weak limit, rather than assuming the contraction theorem cited in Section 3.2. The development also contains reusable results on first-order Taylor bounds, equivalence of the L^2 Poincaré and Rayleigh formulations, reversibility and Dirichlet forms of kernel mixtures, lazification, flow aggregation, spectral-gap upper bounds, and weak-limit stability.

The sole direct external Lean library is mathlib [The mathlib Community, 2020]; its standard dependencies and Lean’s logical foundations are retained.

The complete development was kernel-checked with Lean 4.33.0 and mathlib 4.33.0; the full build and the axiom audit of 266 principal declarations succeeded, with no proof placeholders or project-specific axioms.

6 Numerical experiments

We compare fixed-step MALA and uniformly randomized-step MALA on a perturbed-Gaussian target. The simulation package can be found in a GitHub repository [Qin, 2026].

Let h_0 be a positive number and consider a potential function

$$U_{d,h_0}(x) = \frac{m}{2}x_1^2 + \sum_{i=2}^d \left[\frac{L+m}{4}x_i^2 - \frac{(L-m)h_0}{2} \cos\left(\frac{x_i}{\sqrt{h_0}}\right) \right]. \quad (22)$$

In Section A, perturbed Gaussian distributions with densities proportional to $e^{-U_{d,h}}$ are investigated from a theoretical perspective. As shown in Proposition A.1, $mI_d \preceq \nabla^2 U_{d,h_0}(x) \preceq LI_d$ for every x . Throughout, $m = 0.1$ and $L = 1$, so $\kappa = 10$. This type of construction appears in [Chewi et al., 2021, Lee et al., 2021, Wu et al., 2022], and it serves as a hard target for fixed-step MALA.

For an endpoint $H > 0$, we compare a fixed-step MALA P_h with $h = H$ and the randomized-step $\bar{P}_H$. We report the mean acceptance probability $\mathbb{E}[\alpha_h(X, Y)]$ and the expected squared jumping distance per coordinate,

$$\text{ESJD} = \frac{1}{d} \mathbb{E} \left[\alpha_h(X, Y) \|Y - X\|^2 \right],$$

where X is the current state, and Y is the proposed state. For fixed-step MALA, $Y \mid X, h \sim Q_h(X, \cdot)$ where $h = H$ is fixed. For randomized-step MALA, $Y \mid X, h \sim Q_h(X, \cdot)$ and $h \sim \text{Unif}(0, H)$. We

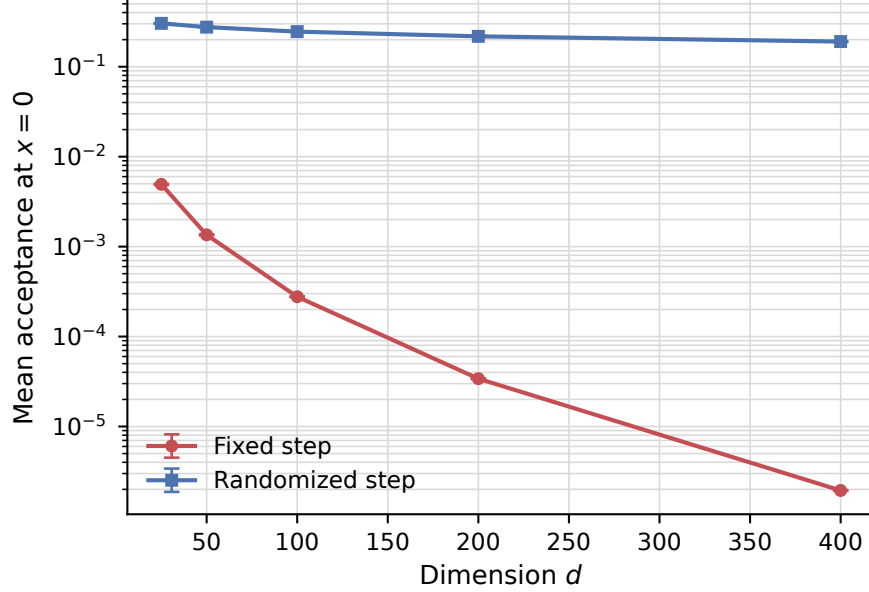


Figure 1: Estimated mean acceptance probability at the origin for the target (22). Fixed-step MALA uses $h = h_0$, while randomized-step MALA uses $h \sim \text{Unif}(0, h_0)$. The vertical axis is logarithmic, and vertical bars show ± 2 estimated standard errors; most are narrower than the markers.

will investigate the setting $X = 0$ as well as $X \sim \pi$. To estimate the mean acceptance rate and the ESJD, we use classical Monte Carlo. That is, independent copies of (X, Y, h) are generated, and expectations under the empirical distribution are used as estimates.

6.1 Acceptance at the origin

For each $d \in \{25, 50, 100, 200, 400\}$, we set

$$h_0 = \frac{4}{L\sqrt{d}}$$

and compare fixed-step MALA with $h = h_0$ against randomized-step MALA with $H = h_0$ in terms of the mean acceptance rate at the origin. As shown in Section A, the origin is where fixed-step MALA struggles to leave. Each estimate uses 80,000 independent proposals.

As Figure 1 shows, the fixed-step acceptance probability decreases from approximately 4.92×10^{-3} at $d = 25$ to 1.94×10^{-6} at $d = 400$. The randomized-step acceptance probability decreases much more slowly, from 0.304 to 0.190. Because rejection leaves a chain initialized at the origin unchanged, the time to its first accepted move is geometric. At $d = 400$, the reciprocal acceptance estimates imply mean waiting times of approximately 5.14×10^5 iterations for fixed-step MALA and 5.25 iterations for randomized-step MALA. The experiment therefore directly displays the sticky behavior behind the fixed-step obstruction: proposals at the oscillation scale h_0 have rapidly deteriorating acceptance, whereas uniform randomization retains a substantial probability of selecting a smaller step.

To continue the origin experiment beyond one step, we next examine what happens after fixed-step MALA eventually escapes the hard point. We run one chain of each type for 40,000 iterations on the $d = 200$ target, starting both chains at the origin and taking $H = h_0$. We record the first coordinate at every iteration.

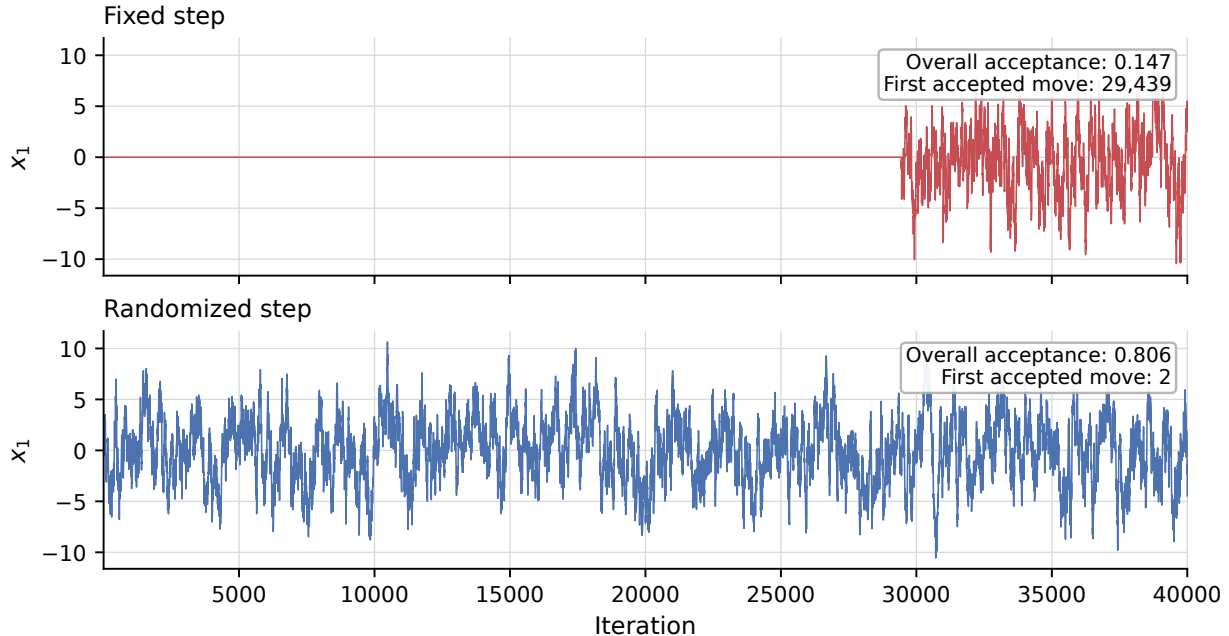


Figure 2: First-coordinate traces continuing the origin experiment for the $d = 200$ hard target, with both chains initialized at the origin and $H = h_0 = 4/\sqrt{200}$. The two panels use common horizontal and vertical scales.

In this realization, the fixed-step chain rejects its first 29,438 proposals and first moves at iteration 29,439. It then makes 5,882 accepted moves by iteration 40,000, so its overall acceptance rate is 0.147. Over the 10,562-iteration interval from its first accepted move through the end of the run, its empirical acceptance rate is 0.557. The trace therefore shows that, once the fixed chain escapes the sticky point, it begins moving on an ordinary scale; the low overall acceptance rate is dominated by the long initial wait. The randomized chain first moves at iteration 2, makes 32,253 accepted moves, and has overall acceptance rate 0.806.

6.2 Stationary acceptance and jumping distance

The preceding experiment deliberately starts at an atypical point. We next compare average one-step behavior under stationarity. We fix $d = 200$ and $h_0 = 4/\sqrt{200}$, and vary H/h_0 from 0.05 to 100, using a fine grid near the efficient range and a logarithmically spaced grid on the over-tuning side.

Exact stationary samples are available because (22) is a product target. The first coordinate is $N(0, m^{-1})$. For each remaining coordinate, we propose $\tilde{X} \sim N(0, 2/(L + m))$ and accept it with probability

$$\exp \left\{ \frac{(L - m)h_0}{2} \left[\cos \left(\frac{\tilde{X}}{\sqrt{h_0}} \right) - 1 \right] \right\}.$$

This is an exact rejection sampler. At every endpoint H and for each method, we use 48,000 independent stationary one-step experiments.

Figure 3 shows the expected efficiency–robustness tradeoff. At $H = h_0$, fixed-step MALA has acceptance 0.559 and ESJD 0.340, while randomized-step MALA has acceptance 0.804 and ESJD 0.216. The largest observed fixed-step ESJD is 0.343, at $H/h_0 \approx 1.21$, whereas the largest observed randomized-step ESJD is 0.263, at $H/h_0 \approx 1.90$. Thus a well-tuned fixed step has the larger peak

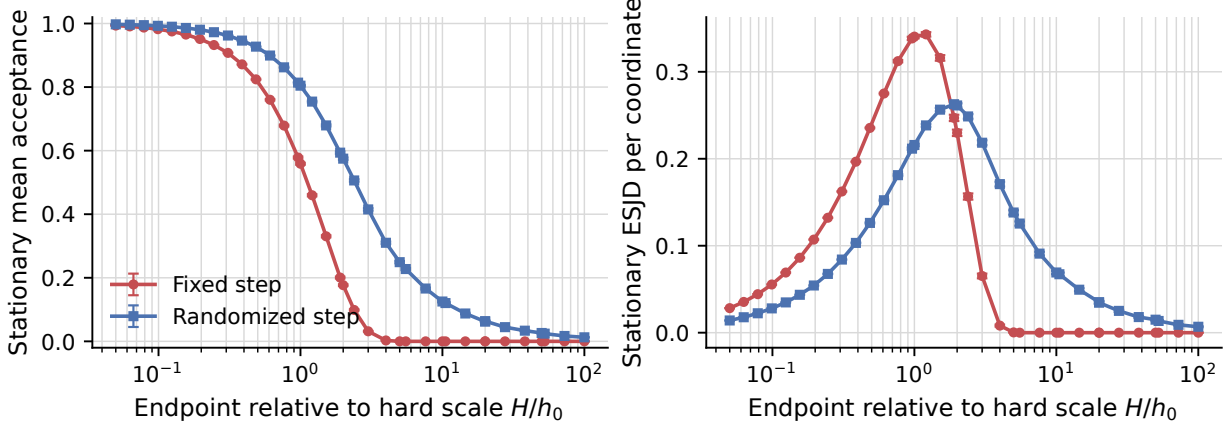


Figure 3: Stationary one-step performance for the $d = 200$ target with $h_0 = 4/\sqrt{200}$. Fixed-step MALA uses $h = H$, while randomized-step MALA uses $h \sim \text{Unif}(0, H)$. Vertical bars show ± 2 estimated standard errors.

ESJD.

Randomization becomes advantageous when H is too large. At $H = 10h_0$, fixed-step acceptance is extremely close to 0, whereas randomized-step MALA retains acceptance 0.125 and ESJD 0.0692. Even at $H = 100h_0$, the randomized method has acceptance 0.0127 and ESJD 0.00681, while the corresponding fixed-step estimates are below floating-point resolution. Randomization does not prevent performance from eventually decreasing as H grows, but it changes the abrupt fixed-step collapse into a much more gradual decline.

The two experiments also expose a distinction relevant to the global spectral gap. When $d = 200$ and $H = h_0$, fixed-step acceptance is about 0.559 under stationarity but only 3.40×10^{-5} at the origin. Thus stationary-average diagnostics largely miss the small sticky region, which leads to the exceedingly small spectral gap of fixed-step MALA, as shown in [Section A](#). Uniform randomization improves acceptance both in that region and under over-tuning, although it does not dominate a well-tuned fixed step in stationary ESJD. These one-step experiments illustrate tuning robustness; they do not estimate the spectral gap itself.

Acknowledgment: The author thanks Guanyang Wang for reading the paper and for providing helpful comments.

Appendices

A Proof of [Proposition 2.3](#)

We follow an existing line of research on hard targets for fixed-step MALA. Perturbed-Gaussian constructions appear in [Chewi et al. \[2021, Theorems 8, 9, and 36\]](#). The product structure combining a Gaussian coordinate with cosine-perturbed Gaussian coordinates appears in [Lee et al. \[2021, Supplement, equation \(20\)\]](#) and [Wu et al. \[2022, Equation \(28\) and Lemma 8\]](#). Below, we use a modified version of the latter construction to obtain the explicit bound in [Proposition 2.3](#).

Proposition A.1. *There are universal constants $c, C > 0$ such that the following holds. Let $d \geq 2$, $0 < m < L$, and $h > 0$. Define*

$$U_{d,h}(x) = \frac{m}{2}x_1^2 + \sum_{i=2}^d \left[\frac{L+m}{4}x_i^2 - \frac{(L-m)h}{2} \cos\left(\frac{x_i}{\sqrt{h}}\right) \right]. \quad (23)$$

Then $U_{d,h}$ is infinitely differentiable and

$$mI_d \preceq \nabla^2 U_{d,h}(x) \preceq LI_d, \quad x \in \mathbb{R}^d. \quad (24)$$

If P_h is the MALA kernel with step size h and target density proportional to $e^{-U_{d,h}}$, then

$$\text{Gap}_R(P_h) \leq C \min \left\{ mh + \frac{(mh)^2}{2}, \exp[-c(d-1) \min\{(L-m)h, 1\}] \right\}. \quad (25)$$

Writing $\kappa = L/m$, this further implies, after possibly changing the universal constants c and C , that

$$\text{Gap}_R(P_h) \leq C \min \left\{ \kappa^{-1}Lh, \exp[-cd \min\{(1-\kappa^{-1})Lh, 1\}] \right\}. \quad (26)$$

Proof. For $i \geq 2$, the i th diagonal entry of the Hessian of $U_{d,h}$ is

$$\frac{L+m}{2} + \frac{L-m}{2} \cos\left(\frac{x_i}{\sqrt{h}}\right),$$

which lies in $[m, L]$. The first diagonal entry equals m , and all off-diagonal entries vanish. This proves (24).

We first establish the local upper bound. The target is a product measure, and its first marginal is $N(0, m^{-1})$. Use $f(x) = x_1$ in the Poincaré quotient. Let X have the target distribution, let $Z \sim N_d(0, I_d)$ be independent of X , and write

$$\tilde{Y} = X - h\nabla U_{d,h}(X) + \sqrt{2h}Z$$

for the MALA proposal. The actual transition either moves to $\tilde{Y}$ or remains at X . Consequently,

$$\begin{aligned} \mathcal{E}_{P_h}(f, f) &\leq \frac{1}{2} \mathbb{E} \left[(-hmX_1 + \sqrt{2h}Z_1)^2 \right] \\ &= \frac{1}{2} (h^2m + 2h), \end{aligned}$$

where we used $\mathbb{E}(X_1^2) = m^{-1}$ and the independence of X_1 and Z_1 . Since $\text{Var}_\pi(f) = m^{-1}$, it follows that

$$\text{Gap}_R(P_h) \leq mh + \frac{(mh)^2}{2}. \quad (27)$$

We next obtain the exponential upper bound by controlling the mean acceptance probability at the origin. Write

$$a = \frac{L+m}{2}, \quad b = \frac{L-m}{2}.$$

At the origin, $\nabla U_{d,h}(0) = 0$, so a proposal has the form

$$Y = \sqrt{2h}Z.$$

For a general differentiable potential U , the log Hastings ratio for a proposal from x to y has the following standard representation; see also [Eberle \[2014, Proposition 1.3, equation \(1.33\)\]](#).

$$\begin{aligned} R_h(x, y) &:= \log \frac{e^{-U(y)} q_h(y, x)}{e^{-U(x)} q_h(x, y)} \\ &= U(x) - U(y) + \frac{1}{2} \langle y - x, \nabla U(x) + \nabla U(y) \rangle \\ &\quad + \frac{h}{4} \left[\|\nabla U(x)\|^2 - \|\nabla U(y)\|^2 \right]. \end{aligned} \tag{28}$$

For $i = 2, \dots, d$, put

$$V_i = \frac{Y_i}{\sqrt{h}} = \sqrt{2} Z_i.$$

Substituting (23) into (28) gives

$$\begin{aligned} R_h(0, Y) &= -\frac{m^2 h^2}{2} Z_1^2 + b h \sum_{i=2}^d A(V_i) \\ &\quad - \frac{h^2}{4} \sum_{i=2}^d [a V_i + b \sin(V_i)]^2, \end{aligned} \tag{29}$$

where

$$A(v) = \cos v - 1 + \frac{v \sin v}{2}.$$

Both the first and the last terms in (29) are nonpositive.

Since $V_i \sim N(0, 2)$,

$$\mathbb{E}(\cos V_i) = e^{-1}, \quad \mathbb{E}(V_i \sin V_i) = 2e^{-1}.$$

The first identity follows from the characteristic function of V_i , and the second follows from Stein's identity. Hence

$$\mathbb{E}[A(V_i)] = \frac{2}{e} - 1 = -a_0, \quad a_0 = 1 - \frac{2}{e} > 0.$$

Moreover,

$$|A(v)| \leq 2 + \frac{|v|}{2}.$$

Thus $A(V_i) - \mathbb{E}[A(V_i)]$ has a universally bounded sub-Gaussian norm; see, for example, [\[Vershynin, 2018, Section 2.5.2\]](#). The concentration inequality for sums of independent sub-Gaussian random variables [\[Vershynin, 2018, Theorem 2.6.3\]](#) therefore yields a universal constant $c_0 > 0$ such that

$$\mathbb{P} \left(\sum_{i=2}^d A(V_i) > -\frac{a_0(d-1)}{2} \right) \leq e^{-c_0(d-1)}. \tag{30}$$

Outside the event in (30),

$$R_h(0, Y) \leq -\frac{a_0}{2} b h (d-1) = -\frac{a_0}{4} (d-1) (L-m) h.$$

Let

$$a_h(x) = \int \alpha_h(x, y) Q_h(x, dy)$$

denote the pointwise acceptance probability. It follows that

$$\begin{aligned} a_h(0) &= \mathbb{E} \left[1 \wedge e^{R_h(0,Y)} \right] \\ &\leq \exp \left[-\frac{a_0}{4}(d-1)(L-m)h \right] + e^{-c_0(d-1)} \\ &\leq 2 \exp [-c_1(d-1) \min\{(L-m)h, 1\}] \end{aligned}$$

for a universal constant $c_1 > 0$.

The function $x \mapsto a_h(x)$ is continuous. Indeed, writing the proposal as

$$x - h \nabla U_{d,h}(x) + \sqrt{2h} Z, \quad Z \sim N_d(0, I_d),$$

the resulting acceptance probability is continuous in x for every fixed Z and is bounded by one. Continuity therefore follows from dominated convergence.

Consequently, there is an open ball S centered at the origin such that

$$0 < \pi(S) \leq \frac{1}{2}, \quad \sup_{x \in S} a_h(x) \leq 4 \exp [-c_1(d-1) \min\{(L-m)h, 1\}].$$

A transition starting in S can leave S only if its proposal is accepted. Hence

$$\mathcal{J}_{P_h}(S, S^c) \leq 4 \exp [-c_1(d-1) \min\{(L-m)h, 1\}] \pi(S).$$

Testing the Poincaré quotient with $\mathbf{1}_S$ gives

$$\begin{aligned} \text{Gap}_R(P_h) &\leq \frac{\mathcal{E}_{P_h}(\mathbf{1}_S, \mathbf{1}_S)}{\text{Var}_\pi(\mathbf{1}_S)} \\ &= \frac{\mathcal{J}_{P_h}(S, S^c)}{\pi(S)[1 - \pi(S)]} \\ &\leq 8 \exp [-c_1(d-1) \min\{(L-m)h, 1\}]. \end{aligned} \tag{31}$$

Combining (27) and (31) proves (25).

Finally, we prove (26). We have

$$L - m = (1 - \kappa^{-1})L, \quad d - 1 \geq \frac{d}{2}.$$

Therefore

$$(d-1) \min\{(L-m)h, 1\} \geq \frac{d}{2} \min\{(1 - \kappa^{-1})Lh, 1\}.$$

If $mh \leq 1$, then

$$mh + \frac{(mh)^2}{2} \leq \frac{3}{2}mh = \frac{3}{2}\kappa^{-1}Lh.$$

If $mh > 1$, then

$$\exp \left[-cd \min\{(1 - \kappa^{-1})Lh, 1\} \right] \leq 1 < mh = \kappa^{-1}Lh,$$

so the exponential bound is the smaller of the two terms in (26). After adjusting the universal constants, this proves (26). $\square$

We can now prove [Proposition 2.3](#).

Proof. By [Proposition A.1](#), there are universal constants $c_0, C_0 > 0$ such that, for every $h > 0$,

$$\inf_{\substack{U \in C^\infty(\mathbb{R}^d) \\ mI_d \preceq \nabla^2 U(x) \preceq LI_d, x \in \mathbb{R}^d}} \text{Gap}_R(P_{U,h}) \leq F(Lh), \quad (32)$$

where

$$F(t) = C_0 \min \left\{ \kappa^{-1}t, \exp \left[-c_0 d \min \{ (1 - \kappa^{-1})t, 1 \} \right] \right\}.$$

Set

$$t_\star = \frac{2 \log(\kappa d)}{c_0(1 - \kappa^{-1})d}.$$

If $0 < t \leq t_\star$, then

$$F(t) \leq C_0 \kappa^{-1}t_\star = \frac{2C_0 \log(\kappa d)}{c_0(1 - \kappa^{-1})\kappa d}.$$

If $t > t_\star$, then the exponential term defining F is nonincreasing in t , and hence

$$\begin{aligned} F(t) &\leq C_0 \exp \left[-c_0 d \min \{ (1 - \kappa^{-1})t_\star, 1 \} \right] \\ &= C_0 \max \left\{ \frac{1}{(\kappa d)^2}, e^{-c_0 d} \right\}. \end{aligned}$$

Consequently,

$$\begin{aligned} \sup_{t>0} F(t) &\leq C_0 \max \left\{ \frac{2 \log(\kappa d)}{c_0(1 - \kappa^{-1})\kappa d}, \frac{1}{(\kappa d)^2}, e^{-c_0 d} \right\} \\ &\leq C_0 \max \left\{ \frac{2 \log(\kappa d)}{c_0(1 - \kappa_0^{-1})\kappa d}, \frac{1}{(\kappa d)^2}, e^{-c_0 d} \right\}. \end{aligned} \quad (33)$$

Because $d \geq 2$ and $\kappa > 1$,

$$\frac{1}{(\kappa d)^2} \leq C \frac{\log(\kappa d)}{\kappa d}$$

for a universal constant C . Absorbing the factors depending only on κ_0 into C , and changing the universal exponential constant if necessary, [\(32\)](#) and [\(33\)](#) prove [\(6\)](#). $\square$

B Proof of [Proposition 3.2](#)

This appendix proves [Proposition 3.2](#). The convexity assumption on U is not needed for [Proposition B.1](#) and [Lemmas B.2](#) to [B.5](#) below. For these results it suffices that $U \in C^1(\mathbb{R}^d)$, that ∇U be globally L -Lipschitz for some $L > 0$, and that $0 < \int_{\mathbb{R}^d} e^{-U(x)} dx < \infty$, with π the normalized measure and $X_0 \sim \pi$. The proofs of the stated results below use only these weaker assumptions. The globally safe assertion of [Proposition 3.2](#), proved at the end of this appendix, continues to use convexity.

The main input is the following stationary estimate for the pointwise rejection function

$$r_h(x) = 1 - \int \alpha_h(x, y) Q_h(x, dy),$$

where

$$\alpha_h(x, y) = 1 \wedge \frac{e^{-U(y)} q_h(y, x)}{e^{-U(x)} q_h(x, y)}, \quad q_h(x, y) = (4\pi h)^{-d/2} \exp \left\{ -\frac{\|y - x + h \nabla U(x)\|^2}{4h} \right\}.$$

Proposition B.1. *There are universal constants $c, C > 0$ such that, for every real $p \geq 1$,*

$$h \leq \frac{c}{L\sqrt{p(d+p)}} \implies \|r_h\|_{L^p(\pi)} \leq CLh\sqrt{p(d+p)}. \quad (34)$$

The proof of [Proposition B.1](#) compares the exact stationary Langevin path with the path whose drift is frozen at its initial position. The endpoint of the latter path is exactly the MALA proposal. Similar diffusion-comparison ideas appear in [Bou-Rabee and Vanden-Eijnden \[2010\]](#), [Bou-Rabee and Hairer \[2013\]](#), [Chewi et al. \[2021\]](#).

The idea behind the proof is roughly as follows. Let $X_0 \sim \pi$, and suppose that, on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$, X_t solves the Langevin equation

$$dX_t = -\nabla U(X_t) dt + \sqrt{2} dB_t, \quad (35)$$

where B_t is a Brownian motion. Standard theory of Langevin diffusion implies $X_h \sim \pi$. Denote by S_h the conditional distribution of X_h given X_0 . We compare $S_h(x, dy)$ to the proposal $Q_h(x, dy)$, which gives the Gaussian law $N_d(x - h\nabla U(x), 2hI_d)$. Define $\nu_h(dx, dy) = \pi(dx)S_h(x, dy)$, $\mu_h(dx, dy) = \pi(dx)Q_h(x, dy)$. Assume that the Radon-Nikodym derivative $F_h = d\mu_h/d\nu_h$ can be found. Since the Langevin diffusion is π -reversible, it holds that ν_h is symmetric, and

$$\frac{d\mu_h^\top}{d\mu_h}(x, y) = \frac{F_h(y, x)}{F_h(x, y)},$$

where $\mu_h^\top(dx, dy) = \mu_h(dy, dx)$. Moreover, it can be argued that, for π -a.e. x , $F_h(x, \cdot) = dQ_h(x, \cdot)/dS_h(x, \cdot)$. Then

$$\begin{aligned} r_h(x) &= \int S_h(x, dy) \left[1 - F_h(x, y) \min \left\{ 1, \frac{F_h(y, x)}{F_h(x, y)} \right\} \right] \\ &\leq \int (|F_h(x, y) - 1| + |F_h(y, x) - 1|) S_h(x, dy). \end{aligned}$$

Using Jensen's inequality and the fact that ν_h is symmetric, one can now show that $\|r_h\|_{L^p(\pi)}$ can be controlled by $2\|F_h - 1\|_{L^p(\nu_h)}$.

To find F_h , let

$$dW_t = dB_t - \theta_t dt, \quad \theta_t = \frac{\nabla U(X_t) - \nabla U(X_0)}{\sqrt{2}}$$

Then X_t has the frozen drift representation $dX_t = -\nabla U(X_0)dt + \sqrt{2}dW_t$. Suppose that one can find an alternative probability measure $\mathbb{P}_h^{\text{fr}}$ defined by a Radon-Nikodym derivative $D_h = d\mathbb{P}_h^{\text{fr}}/d\mathbb{P}|_{\mathcal{F}_h}$ such that, under $\mathbb{P}_h^{\text{fr}}$, W_t is a Brownian motion with respect to $(\mathcal{F}_t)_{0 \leq t \leq h}$. Then, under this new probability measure, (X_0, X_h) is distributed as μ_h . In particular, with expectations defined in terms of $\mathbb{P}$, $F_h(X_0, X_h) = \mathbb{E}[D_h | \sigma(X_0, X_h)]$, so conditional Jensen's inequality would imply that $\|F_h - 1\|_{L^p(\nu_h)} \leq \|D_h - 1\|_{L^p(\mathbb{P})}$. As stated in [Lemma B.4](#), one can construct the desired measure $\mathbb{P}_h^{\text{fr}}$ by letting D_h be defined through the stochastic exponential

$$D_t = \exp \left(\int_0^t \langle \theta_s, dB_s \rangle - \frac{1}{2} V_t \right), \quad V_t = \int_0^t \|\theta_s\|^2 ds,$$

provided that $(D_t)_{0 \leq t \leq h}$ is a true martingale. This is a consequence of Girsanov's theorem [see, e.g., [Karatzas and Shreve, 1991](#), [Revuz and Yor, 1999](#)]. Therefore, to prove [Proposition B.1](#), it comes down to two tasks: showing that D_t is a true martingale, and that $\|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}$ when $h \leq c/(L\sqrt{p(d+p)})$, where c, C are appropriate universal constants.

To show that D_t is a martingale, it suffices to establish Novikov's criterion: $\mathbb{E}(e^{V_h/2}) < \infty$. Since ∇U is L -Lipschitz, one can bound V_h by $L^2 J_h/2$, where $J_h = \int_0^h \|X_t - X_0\|^2 dt$. This prompts us to bound $\mathbb{E}(e^{\lambda J_h})$ for some suitable $\lambda \geq 0$ in [Lemma B.3](#).

We obtain the upper bound $\|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}$ for $h \leq c/(L\sqrt{p(d+p)})$ in [Lemma B.5](#). In the proof of [Lemma B.5](#), we apply Itô's formula, Burkholder-Davis-Gundy (BDG) inequality, and Doob's maximal inequality to obtain what is loosely

$$\|D_h - 1\|_{L^p(\mathbb{P})} \leq C' \sqrt{p} \|V_h\|_{L^p(\mathbb{P})}^{1/2} \leq 2^{-1/2} C' L \sqrt{p} \|J_h\|_{L^p(\mathbb{P})}^{1/2},$$

where C' is universal. This prompts us to study $\|J_h\|_{L^p(\mathbb{P})}$ and obtain an upper bound that is of order $h^2(d+p)$ in [Lemma B.3](#).

To establish the bounds in [Lemma B.3](#), we first control $X_t - X_0$ using stationary reversibility in [Lemma B.2](#), without explicitly solving the Langevin equation.

Once [Proposition B.1](#) is established, we can utilize elementary techniques in the MCMC literature [see, e.g., [Wu et al., 2022](#), proof of Theorem 3] to establish [Proposition 3.2](#).

Attribution within this appendix. The estimate in [Lemma B.2](#) is an elementary consequence of the Lyons–Zheng forward–backward decomposition [[Lyons and Zheng, 1988](#), Section 1, especially equation (1.7)]; see also [Fukushima et al. \[2011, Theorem 5.7.1\]](#). For the Langevin diffusion considered here, the required decomposition follows directly from the stochastic differential equation and stationary reversibility, as shown below. [Lemma B.3](#) then follows from that estimate, Gaussian randomization, and Jensen's inequality. The Gaussian-randomization step is the identity-matrix specialization of the argument used for sub-Gaussian quadratic forms by [Hsu et al. \[2012, Theorem 1 and Remark 2\]](#). [Lemma B.5](#) combines standard results on stochastic exponentials, Novikov's criterion, Girsanov's theorem, stopping of stochastic integrals, Doob's inequality, and the quantitative BDG inequality, with the problem-specific estimate $V_h \leq L^2 J_h/2$. General reverse-Hölder theory for continuous stochastic exponentials is developed by [Kazamaki \[1994\]](#), but the dimension- and moment-explicit calculation needed here is supplied in full. The passage from the path likelihood to the rejection probability uses the standard conditional-likelihood identity and the Metropolis–Hastings meet identity [[Kallenberg, 2021](#), [Tierney, 1998](#), [Billera and Diaconis, 2001](#)]. Thus [Proposition B.1](#) combines these ingredients for the present MALA problem.

B.1 Controlling Langevin increments

The goal of this subsection is to control the magnitude of $X_t - X_0$, particularly the time integral $\int_0^h \|X_t - X_0\|^2 dt$, where X_t satisfies (35).

We begin by fixing the continuous-time framework used throughout this appendix. All random variables are defined on a filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$$

satisfying the usual completeness and right-continuity conditions. A standard d -dimensional Brownian motion relative to this filtration is an adapted continuous process B with $B_0 = 0$ such that, for $0 \leq s < t$, the increment $B_t - B_s$ has law $N_d(0, (t-s)I_d)$ and is independent of $\mathcal{F}_s$. This last independence property will be used explicitly in [Section B.2](#). Basic references for filtered probability spaces, Brownian motion, Itô integration, and stochastic differential equations are [Karatzas and Shreve \[1991, Chapters 3 and 5\]](#), [Revuz and Yor \[1999, Chapters IV and IX\]](#), and [Kallenberg \[2021\]](#).

Because ∇U is globally Lipschitz, for every $x \in \mathbb{R}^d$ the Langevin equation has a unique nonexplosive strong solution $(X_t^x)_{t \geq 0}$ with $X_0^x = x$; see, for example, [Karatzas and Shreve \[1991\]](#),

Chapter 5]. For a bounded measurable function f , define the transition semigroup by

$$S_t f(x) = \mathbb{E}[f(X_t^x)].$$

Now let X_0 be $\mathcal{F}_0$ -measurable and independent of B , with law π , and let $(X_t)_{t \geq 0}$ be the corresponding solution of

$$dX_t = -\nabla U(X_t) dt + \sqrt{2} dB_t, \quad X_0 \sim \pi. \quad (36)$$

The differential notation means the integral identity

$$X_t = X_0 - \int_0^t \nabla U(X_s) ds + \sqrt{2} B_t, \quad t \geq 0. \quad (37)$$

Under the assumptions stated at the beginning of this appendix, the Langevin semigroup is symmetric with respect to $\pi(dx) \propto e^{-U(x)} dx$:

$$\int f S_t g d\pi = \int g S_t f d\pi$$

for bounded measurable f, g and $t \geq 0$; see [Durmus and Moulines \[2017, Section 2\]](#) and [Fukushima et al. \[2011, Chapters 1 and 7\]](#). Since $S_t 1 = 1$, taking $f = 1$ shows that π is invariant. Consequently, the process started from $X_0 \sim \pi$ is stationary and reversible.

Lemma B.2. *For every $v \in \mathbb{R}^d$, $\lambda \in \mathbb{R}$, and $t \geq 0$,*

$$\mathbb{E}[\exp(\lambda \langle v, X_t - X_0 \rangle)] \leq \exp(\lambda^2 t \|v\|^2). \quad (38)$$

Proof. The case $t = 0$ is immediate. Fix $t > 0$. Stationarity and reversibility give

$$(X_s)_{0 \leq s \leq t} \stackrel{d}{=} (X_{t-s})_{0 \leq s \leq t}$$

as random elements of $C([0, t]; \mathbb{R}^d)$. Set

$$Z_t = X_0 - X_t + \int_0^t \nabla U(X_s) ds.$$

Reversing the path interchanges X_0 and X_t , while leaving the time integral unchanged, since

$$\int_0^t \nabla U(X_{t-s}) ds = \int_0^t \nabla U(X_s) ds.$$

The equality of path laws and (37) therefore give

$$Z_t \stackrel{d}{=} X_t - X_0 + \int_0^t \nabla U(X_s) ds = \sqrt{2} B_t \sim N_d(0, 2tI_d).$$

Moreover, subtracting the definition of Z_t from the preceding expression for $\sqrt{2} B_t$ yields

$$X_t - X_0 = \frac{\sqrt{2} B_t - Z_t}{2}.$$

Thus the drift integral cancels when the forward and reversed expressions are subtracted.

By convexity of the exponential function,

$$\begin{aligned}\mathbb{E}[\exp(\lambda \langle v, X_t - X_0 \rangle)] &= \mathbb{E} \left[\exp \left(\frac{\lambda}{2} \langle v, \sqrt{2} B_t - Z_t \rangle \right) \right] \\ &\leq \frac{1}{2} \mathbb{E}[\exp(\sqrt{2} \lambda \langle v, B_t \rangle)] + \frac{1}{2} \mathbb{E}[\exp(-\lambda \langle v, Z_t \rangle)] \\ &= \exp(\lambda^2 t \|v\|^2).\end{aligned}$$

The last equality uses the Gaussian moment-generating formula. No independence between B_t and Z_t is required, and the finite Gaussian exponential moments justify the inequality without any preliminary integrability assumption on its left-hand side. $\square$

The next proof uses a standard Gaussian-randomization argument. In a more general matrix form, it is the argument used by [Hsu et al. \[2012, Theorem 1 and Remark 2\]](#) to pass from linear sub-Gaussian moment bounds to quadratic-form bounds. We include the identity-matrix calculation because the exact constant in (39) is useful below.

Lemma B.3. *For every $t > 0$ and $0 \leq a < 1/(4t)$,*

$$\mathbb{E}[\exp(a \|X_t - X_0\|^2)] \leq (1 - 4at)^{-d/2}. \quad (39)$$

If

$$J_h = \int_0^h \|X_t - X_0\|^2 dt,$$

then, for $0 \leq \lambda < 1/(4h^2)$,

$$\mathbb{E}(e^{\lambda J_h}) \leq (1 - 4\lambda h^2)^{-d/2}. \quad (40)$$

Consequently, for every real $p \geq 1$,

$$\|J_h\|_{L^p(\mathbb{P})} \leq Ch^2(d + p), \quad (41)$$

where C is universal.

Proof. Enlarge the probability space, if necessary, so that it also carries a random vector $G \sim N_d(0, I_d)$ independent of the entire Langevin path $(X_s)_{s \geq 0}$. For a fixed vector $z \in \mathbb{R}^d$, the scalar $\langle G, z \rangle$ is Gaussian with mean zero and variance $\|z\|^2$. The Gaussian moment-generating formula therefore gives

$$\mathbb{E}(e^{\sqrt{2a}\langle G, z \rangle}) = e^{a\|z\|^2}.$$

Apply this identity with the random vector $z = X_t - X_0$. More precisely, conditioning first on the Langevin path gives

$$\mathbb{E}(e^{a\|X_t - X_0\|^2}) = \mathbb{E} \left\{ \mathbb{E} \left[e^{\sqrt{2a}\langle G, X_t - X_0 \rangle} \mid \sigma(X_s : s \geq 0) \right] \right\}.$$

The integrand is nonnegative, so Tonelli's theorem allows us to reverse the order in which the path and G are averaged. Conditional on G , the vector G is fixed, and [Lemma B.2](#), with $v = G$ and $\lambda = \sqrt{2a}$, gives

$$\mathbb{E}(e^{\sqrt{2a}\langle G, X_t - X_0 \rangle} \mid G) \leq e^{2at\|G\|^2}.$$

Consequently,

$$\begin{aligned}\mathbb{E}(e^{a\|X_t - X_0\|^2}) &\leq \mathbb{E}(e^{2at\|G\|^2}) \\ &= (1 - 4at)^{-d/2},\end{aligned}$$

where the last equality is the moment-generating function of a χ_d^2 random variable and requires $4at < 1$. This proves (39).

We next pass from one increment to its time integral. For each fixed sample path, Jensen's inequality for the convex function $x \mapsto e^x$, applied to the uniform probability measure dt/h on $[0, h]$, gives

$$\begin{aligned} e^{\lambda J_h} &= \exp \left(\frac{1}{h} \int_0^h \lambda h \|X_t - X_0\|^2 dt \right) \\ &\leq \frac{1}{h} \int_0^h e^{\lambda h \|X_t - X_0\|^2} dt. \end{aligned}$$

Take expectations and use Tonelli's theorem, followed by (39) with $a = \lambda h$:

$$\begin{aligned} \mathbb{E}(e^{\lambda J_h}) &\leq \frac{1}{h} \int_0^h \mathbb{E}(e^{\lambda h \|X_t - X_0\|^2}) dt \\ &\leq \frac{1}{h} \int_0^h (1 - 4\lambda h t)^{-d/2} dt \\ &\leq (1 - 4\lambda h^2)^{-d/2}. \end{aligned}$$

This proves (40).

It remains to derive the L^p estimate from this exponential-tail bound. Put $\lambda = (8h^2)^{-1}$ in (40). Markov's inequality gives, for every $s \geq 0$,

$$\mathbb{P}(J_h \geq 4dh^2 \log 2 + 8h^2 s) \leq e^{-s}.$$

Define

$$Y = \frac{(J_h - 4dh^2 \log 2)_+}{8h^2}.$$

Then $\mathbb{P}(Y > s) \leq e^{-s}$. For every real $p \geq 1$, the layer-cake formula for moments gives

$$\begin{aligned} \mathbb{E}(Y^p) &= p \int_0^\infty s^{p-1} \mathbb{P}(Y > s) ds \\ &\leq p \int_0^\infty s^{p-1} e^{-s} ds = \Gamma(p+1). \end{aligned}$$

Stirling's bound implies $\Gamma(p+1)^{1/p} \leq Cp$ for all $p \geq 1$. Hence, by the Minkowski inequality,

$$\|J_h\|_{L^p(\mathbb{P})} \leq 4dh^2 \log 2 + 8h^2 \|Y\|_{L^p(\mathbb{P})} \leq Ch^2(d+p),$$

which proves (41). For the tail-integration identity and the Gamma-function estimate, see [Boucheron et al. \[2013, Chapter 2\]](#). $\square$

B.2 Girsanov comparison with frozen drift

We compare two laws on the same path space over $[0, h]$. Under the probability measure $\mathbb{P}$, the path satisfies the exact Langevin equation (36). We shall construct a second measure under which the coordinate process has drift frozen at the initial value $-\nabla U(X_0)$.

We recall the continuous-martingale facts used below. If H_t is a predictable $\mathbb{R}^d$ -valued process with $\int_0^h \|H_t\|^2 dt < \infty$ almost surely, then $\int_0^t \langle H_s, dB_s \rangle$ is a continuous local martingale. Its quadratic variation is

$$\left[\int_0^\cdot \langle H_s, dB_s \rangle \right]_t = \int_0^t \|H_s\|^2 ds.$$

For a continuous local martingale N with $N_0 = 0$, the stochastic exponential is

$$\mathcal{E}(N)_t = \exp(N_t - [N]_t/2).$$

It is always a nonnegative local martingale, but it need not have expectation one. Novikov's criterion states that $\mathbb{E} \exp([N]_h/2) < \infty$ is sufficient for $(\mathcal{E}(N)_t)_{0 \leq t \leq h}$ to be a true mean-one martingale. These statements, together with Itô's formula and the stopping arguments used below, are standard; see Karatzas and Shreve [1991, Chapter 3], Revuz and Yor [1999, Chapters IV and VIII], or Kallenberg [2021].

Define

$$\theta_t = \frac{\nabla U(X_t) - \nabla U(X_0)}{\sqrt{2}}, \quad V_t = \int_0^t \|\theta_s\|^2 ds, \quad M_t = \int_0^t \langle \theta_s, dB_s \rangle.$$

The process θ_t is continuous and adapted, hence predictable. Since X has continuous paths, $\int_0^h \|\theta_s\|^2 ds < \infty$ almost surely on every finite interval, so the Itô integral M_t is well defined as a continuous local $\mathbb{P}$ -martingale. Its quadratic variation is $[M]_t = V_t$. The Lipschitz property of ∇U and Lemma B.3 give

$$V_h \leq \frac{L^2}{2} J_h, \quad \|V_h\|_{L^p(\mathbb{P})} \leq CL^2 h^2 (d+p) \text{ whenever } p \geq 1. \quad (42)$$

Define the stochastic exponential

$$D_t = \exp\left(M_t - \frac{1}{2} V_t\right) = \mathcal{E}(M)_t.$$

The following lemma collects the change-of-measure construction that will be used in the proof of Proposition B.1.

Lemma B.4. *Assume that $h > 0$ and $Lh < 1$. Then $(D_t)_{0 \leq t \leq h}$ is a true mean-one $\mathbb{P}$ -martingale, and*

$$\left. \frac{d\mathbb{P}_h^{\text{fr}}}{d\mathbb{P}} \right|_{\mathcal{F}_h} = D_h \quad (43)$$

defines a probability measure $\mathbb{P}_h^{\text{fr}}$ on $(\Omega, \mathcal{F}_h)$. Under $\mathbb{P}_h^{\text{fr}}$, the process

$$W_t = B_t - \int_0^t \theta_s ds, \quad 0 \leq t \leq h,$$

is a Brownian motion relative to $(\mathcal{F}_t)_{0 \leq t \leq h}$, and

$$X_t = X_0 - t \nabla U(X_0) + \sqrt{2} W_t, \quad 0 \leq t \leq h. \quad (44)$$

Moreover, under $\mathbb{P}_h^{\text{fr}}$, $X_0 \sim \pi$, and a version of the conditional law of X_h given $X_0 = x$ is $Q_h(x, \cdot)$. It follows that, for all Borel sets $A, B \subseteq \mathbb{R}^d$,

$$\mathbb{P}_h^{\text{fr}}(X_0 \in A, X_h \in B) = \int_A Q_h(x, B) \pi(dx). \quad (45)$$

Proof. By (40) and (42),

$$\begin{aligned} \mathbb{E} [\exp([M]_h/2)] &= \mathbb{E} [\exp(V_h/2)] \\ &\leq \mathbb{E} [\exp(L^2 J_h/4)] \end{aligned}$$

$$\leq (1 - L^2 h^2)^{-d/2} < \infty,$$

where the last inequality uses $Lh < 1$. Novikov's criterion therefore shows that D is a true mean-one $\mathbb{P}$ -martingale. In particular,

$$\mathbb{E}(D_t \mid \mathcal{F}_s) = D_s, \quad 0 \leq s \leq t \leq h, \quad (46)$$

and (43) defines a probability measure.

Because D is a true martingale, Girsanov's theorem gives the asserted Brownian motion W_t ; see Karatzas and Shreve [1991, Section 3.5] or Revuz and Yor [1999, Chapter VIII, Section 1]. Substituting

$$B_t = W_t + \int_0^t \theta_s \, ds$$

into (37) yields

$$\begin{aligned} X_t &= X_0 - \int_0^t \nabla U(X_s) \, ds + \sqrt{2} W_t + \sqrt{2} \int_0^t \theta_s \, ds \\ &= X_0 - t \nabla U(X_0) + \sqrt{2} W_t, \end{aligned}$$

which proves (44).

For every Borel set $A \subseteq \mathbb{R}^d$,

$$\begin{aligned} \mathbb{P}_h^{\text{fr}}(X_0 \in A) &= \mathbb{E} \left(D_h \mathbf{1}_{\{X_0 \in A\}} \right) \\ &= \mathbb{E} \left[\mathbb{E}(D_h \mid \mathcal{F}_0) \mathbf{1}_{\{X_0 \in A\}} \right] \\ &= \mathbb{P}(X_0 \in A) = \pi(A), \end{aligned}$$

where (46) gives $\mathbb{E}[D_h \mid \mathcal{F}_0] = D_0 = 1$.

Under $\mathbb{P}_h^{\text{fr}}$, the increment $W_h - W_0 = W_h$ is independent of $\mathcal{F}_0$ and has law $N_d(0, hI_d)$. Since X_0 is $\mathcal{F}_0$ -measurable, (44) gives, conditionally on $X_0 = x$,

$$X_h = x - h \nabla U(x) + \sqrt{2h} Z, \quad Z \sim N_d(0, I_d).$$

This is precisely the kernel $Q_h(x, \cdot)$. Combining this conditional law with $X_0 \sim \pi$ proves (45). $\square$

We next quantify how close the path likelihood D_h is to one. General reverse-Hölder and moment estimates for continuous stochastic exponentials form a classical theory; see, for example, Kazamaki [1994, Chapters 1 and 3]. We give a direct localized calculation to track explicitly the dependence on d, p, L , and h and to exploit the problem-specific bound $V_h \leq L^2 J_h / 2$.

Lemma B.5. *There are universal constants $c, C > 0$ such that, for every $h > 0$ and every real $p \geq 1$, the conditions*

$$Lh < 1, \quad h \leq \frac{c}{L\sqrt{p(d+p)}} \quad (47)$$

imply

$$\|D_h\|_{L^{2p}(\mathbb{P})} \leq C, \quad \|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}. \quad (48)$$

Proof. Stopping conventions. We use standard process-level stopping identities. If N_t is a continuous local martingale and τ is a stopping time, then, up to indistinguishability,

$$N_t^\tau := N_{t \wedge \tau}, \quad [N^\tau]_t = [N]_{t \wedge \tau}. \quad (49)$$

Likewise, for a predictable integrand H_t ,

$$\left(\int_0^\cdot \langle H_s, dB_s \rangle \right)_{t \wedge \tau} = \int_0^t \mathbf{1}_{\{s \leq \tau\}} \langle H_s, dB_s \rangle. \quad (50)$$

These are identities of stopped processes and therefore hold for all t simultaneously outside a single null set; continuity then permits evaluation at any bounded random time. Purely algebraic identities among continuous processes may be evaluated at $t \wedge \tau$ for the same reason, and inequalities such as $V_{t \wedge \tau} \leq V_t$ follow pathwise from monotonicity. By contrast, an estimate obtained only after taking expectations at each fixed time cannot generally be evaluated at a stopping time without a separate optional-stopping argument. No such step is used here. See [Revuz and Yor \[1999, Chapter IV, Sections 1–2\]](#), [Karatzas and Shreve \[1991, Chapter 3\]](#), or [Protter \[2005, Chapter II\]](#) for stopping of continuous local martingales, stochastic integrals, and quadratic variation.

Step 1: a uniform L^{2p} bound for D_h . For $n \geq 1$, let

$$\tau_n = \inf\{t \in [0, h] : V_t \geq n\}, \quad \inf \emptyset = \infty.$$

Because V is continuous and adapted, τ_n is a stopping time. Set $M_t^{(n)} = M_{t \wedge \tau_n}$. By (49),

$$[M^{(n)}]_t = [M]_{t \wedge \tau_n} = V_{t \wedge \tau_n}.$$

Consequently, the quadratic variation of the continuous local martingale $4pM^{(n)}$ is

$$[4pM^{(n)}]_t = 16p^2 V_{t \wedge \tau_n} \leq 16p^2 n, \quad 0 \leq t \leq h.$$

Consider its stochastic exponential

$$\Lambda_t^{(2p,n)} = \exp\left(4pM_{t \wedge \tau_n} - 8p^2 V_{t \wedge \tau_n}\right).$$

Novikov's criterion applies because

$$\mathbb{E} \left[\exp \left(\frac{1}{2} [4pM^{(n)}]_h \right) \right] \leq e^{8p^2 n} < \infty.$$

Hence $\Lambda^{(2p,n)}$ is a true mean-one martingale, and

$$\mathbb{E} \left[\Lambda_h^{(2p,n)} \right] = 1. \quad (51)$$

The definitions of D and $\Lambda^{(2p,n)}$ give the exact algebraic identity

$$D_{h \wedge \tau_n}^{2p} = [\Lambda_h^{(2p,n)}]^{1/2} e^{(4p^2 - p)V_{h \wedge \tau_n}}.$$

Cauchy–Schwarz, (51), and the inequalities $V_{h \wedge \tau_n} \leq V_h$ and $8p^2 - 2p \geq 0$, valid for $p \geq 1$, yield

$$\begin{aligned} \mathbb{E} \left[D_{h \wedge \tau_n}^{2p} \right] &\leq \left\{ \mathbb{E} [\Lambda_h^{(2p,n)}] \right\}^{1/2} \left\{ \mathbb{E} \left[e^{(8p^2 - 2p)V_{h \wedge \tau_n}} \right] \right\}^{1/2} \\ &\leq \left\{ \mathbb{E} \left[e^{(8p^2 - 2p)V_h} \right] \right\}^{1/2}. \end{aligned}$$

Since $V_h \leq L^2 J_h / 2$, as noted in (42), equation (40) gives

$$\mathbb{E} \left[D_{h \wedge \tau_n}^{2p} \right] \leq (1 - u)^{-d/4}, \quad u = 4p(4p - 1)L^2 h^2,$$

provided $u < 1$. By (47),

$$u \leq 16p^2 L^2 h^2 \leq 16c^2 \frac{p}{d+p} \leq 16c^2.$$

Choose the universal constant $c > 0$ in the statement so that $16c^2 \leq 1/2$. It follows that $0 \leq u \leq 1/2$, and hence

$$\log \left(\frac{1}{1-u} \right) \leq 2u.$$

Therefore

$$\begin{aligned} \log \|D_{h \wedge \tau_n}\|_{L^{2p}(\mathbb{P})} &= \frac{1}{2p} \log \left(\mathbb{E} \left[D_{h \wedge \tau_n}^{2p} \right] \right) \\ &\leq \frac{d}{8p} \log \left(\frac{1}{1-u} \right) \\ &\leq \frac{du}{4p} \\ &\leq 4dp L^2 h^2 \\ &\leq 4c^2 \frac{d}{d+p} \\ &\leq 4c^2. \end{aligned}$$

Thus

$$\|D_{h \wedge \tau_n}\|_{L^{2p}(\mathbb{P})} \leq e^{4c^2} \quad (52)$$

uniformly in n . Finally, $V_h < \infty$ almost surely. For every sample path, once $n > V_h$, one has $\tau_n = \infty$ and hence $D_{h \wedge \tau_n} = D_h$. Fatou's lemma applied to the $(2p)$ -th powers in (52) gives

$$\|D_h\|_{L^{2p}(\mathbb{P})} \leq e^{4c^2}. \quad (53)$$

This proves the first estimate in (48).

Step 2: representation of $D_h - 1$ as a stochastic integral. Let

$$Y_t = M_t - \frac{1}{2}V_t, \quad D_t = e^{Y_t}.$$

Since M is a continuous local $\mathbb{P}$ -martingale and V is continuous and of finite variation, Y is a continuous semimartingale with quadratic variation $[Y]_t = [M]_t = V_t$. Applying Itô's formula for continuous semimartingales [Karatzas and Shreve, 1991, Section 3.3.A, Theorem 3.3] to $y \mapsto e^y$ gives

$$\begin{aligned} dD_t &= D_t dY_t + \frac{1}{2} D_t d[Y]_t \\ &= D_t \left(dM_t - \frac{1}{2} dV_t \right) + \frac{1}{2} D_t dV_t \\ &= D_t dM_t \\ &= D_t \langle \theta_t, dB_t \rangle. \end{aligned}$$

Integrating this identity from zero to t gives the local-martingale representation

$$D_t - 1 = \int_0^t D_s \langle \theta_s, dB_s \rangle. \quad (54)$$

See [Karatzas and Shreve \[1991, Chapter 3\]](#) or [Revuz and Yor \[1999, Chapter IV\]](#) for Itô's formula and the quadratic variation rules used here.

Step 3: application of BDG to an explicitly stopped martingale. For $n \geq 2$, define

$$\sigma_n = \tau_n \wedge \inf\{t \in [0, h] : D_t \geq n\}, \quad \inf \emptyset = \infty.$$

Both V_t and D_t are continuous and adapted, so σ_n is a stopping time. Stopping the process identity (54) and using (50) gives, for every deterministic $t \in [0, h]$,

$$\begin{aligned} R_t^{(n)} &:= D_{t \wedge \sigma_n} - 1 \\ &= \int_0^{t \wedge \sigma_n} D_s \langle \theta_s, dB_s \rangle \\ &= \int_0^t \mathbf{1}_{\{s \leq \sigma_n\}} D_s \langle \theta_s, dB_s \rangle. \end{aligned}$$

Thus $R^{(n)}$ is a square-integrable continuous martingale. Indeed, its quadratic variation is

$$[R^{(n)}]_h = \int_0^h \mathbf{1}_{\{s \leq \sigma_n\}} D_s^2 \|\theta_s\|^2 ds = \int_0^{h \wedge \sigma_n} D_s^2 \|\theta_s\|^2 ds \leq n^2 V_{h \wedge \tau_n} \leq n^3.$$

The possible distinction between $s < \sigma_n$ and $s \leq \sigma_n$ is immaterial for the Lebesgue and Itô integrals.

Apply the upper inequality in [Ren \[2008, Theorem 2, equation \(23\)\]](#), valid for every real $p \geq 1$, to the stopped continuous martingale $R^{(n)}$ at time h . This yields

$$\|R_h^{(n)}\|_{L^p(\mathbb{P})} \leq \left\| \sup_{0 \leq t \leq h} |R_t^{(n)}| \right\|_{L^p(\mathbb{P})} \leq 2\sqrt{2p} \| [R^{(n)}]_h^{1/2} \|_{L^p(\mathbb{P})}. \quad (55)$$

Pathwise,

$$[R^{(n)}]_h^{1/2} \leq \left(\sup_{0 \leq s \leq h} D_s \right) V_h^{1/2}.$$

Hölder's inequality with exponents two and two therefore gives

$$\begin{aligned} \|[R^{(n)}]_h^{1/2}\|_{L^p(\mathbb{P})} &\leq \left\| \sup_{0 \leq s \leq h} D_s \right\|_{L^{2p}(\mathbb{P})} \|V_h^{1/2}\|_{L^{2p}(\mathbb{P})} \\ &= \left\| \sup_{0 \leq s \leq h} D_s \right\|_{L^{2p}(\mathbb{P})} \|V_h\|_{L^p(\mathbb{P})}^{1/2}. \end{aligned} \quad (56)$$

By [Lemma B.4](#), since $Lh < 1$, $(D_t)_{0 \leq t \leq h}$ is a nonnegative true martingale. Conditional Jensen's inequality gives

$$\sup_{0 \leq s \leq h} \|D_s\|_{L^{2p}(\mathbb{P})} = \|D_h\|_{L^{2p}(\mathbb{P})}.$$

Since $2p > 1$, Doob's maximal inequality [[Revuz and Yor, 1999, Chapter II, Theorem 1.7](#)] gives

$$\begin{aligned} \left\| \sup_{0 \leq s \leq h} D_s \right\|_{L^{2p}(\mathbb{P})} &\leq \frac{2p}{2p-1} \sup_{0 \leq s \leq h} \|D_s\|_{L^{2p}(\mathbb{P})} \\ &= \frac{2p}{2p-1} \|D_h\|_{L^{2p}(\mathbb{P})} \leq 2e^{4c^2}, \end{aligned}$$

where the last inequality uses (53) and $2p/(2p-1) \leq 2$ for $p \geq 1$. Moreover, (42) gives

$$\|V_h\|_{L^p(\mathbb{P})}^{1/2} \leq CLh\sqrt{d+p}.$$

Substituting these bounds into (55)–(56) gives, uniformly in n ,

$$\|D_{h \wedge \sigma_n} - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}. \quad (57)$$

To remove the stopping, note that the continuous functions $t \mapsto V_t$ and $t \mapsto D_t$ are bounded on the compact interval $[0, h]$ for every sample path. Hence $\sigma_n > h$ for all sufficiently large n , path by path, and

$$D_{h \wedge \sigma_n} - 1 \longrightarrow D_h - 1 \quad \text{almost surely.}$$

Fatou's lemma applied to (57) proves

$$\|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}.$$

This is the second estimate in (48) and completes the proof. $\square$

B.3 From path likelihood to MALA rejection

Proof of Proposition B.1. Choose the universal constant c in Proposition B.1 no larger than the constant c in Lemma B.5 and sufficiently small so that the step-size condition $h \leq c/[L\sqrt{p(d+p)}]$ implies $Lh < 1$. This will allow us to apply Lemmas B.4 and B.5.

Let S_h be the exact Langevin transition kernel. Define two endpoint probability measures on $\mathbb{R}^d \times \mathbb{R}^d$:

$$\nu_h(dx, dy) = \pi(dx)S_h(x, dy), \quad \mu_h(dx, dy) = \pi(dx)Q_h(x, dy).$$

The measure ν_h is the law of (X_0, X_h) under the exact path law $\mathbb{P}$. By Lemma B.4, μ_h is the law of (X_0, X_h) under $\mathbb{P}_h^{\text{fr}}$. Moreover, (43) gives

$$\frac{d\mathbb{P}_h^{\text{fr}}}{d\mathbb{P}} = D_h \quad \text{on } \mathcal{F}_h,$$

so $\mu_h \ll \nu_h$. Let

$$F_h = \frac{d\mu_h}{d\nu_h}.$$

We next pass from the full-path likelihood D_h to the endpoint likelihood F_h . By the Doob–Dynkin lemma, $\mathbb{E}[D_h \mid \sigma(X_0, X_h)]$ has a Borel-measurable representation as a function of the two endpoints; this function is a version of F_h . Indeed, for every bounded Borel function $\varphi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\begin{aligned} \int \varphi(x, y) \mu_h(dx, dy) &= \mathbb{E}_{\mathbb{P}_h^{\text{fr}}}[\varphi(X_0, X_h)] \\ &= \mathbb{E}[D_h \varphi(X_0, X_h)] \\ &= \mathbb{E}\{\mathbb{E}[D_h \mid \sigma(X_0, X_h)]\varphi(X_0, X_h)\}. \end{aligned}$$

This is exactly the defining Radon–Nikodym identity. Thus

$$F_h(X_0, X_h) = \mathbb{E}[D_h \mid \sigma(X_0, X_h)] \quad \mathbb{P}\text{-almost surely.} \quad (58)$$

This is the usual principle that the likelihood ratio of an observed statistic is the conditional expectation of the full-data likelihood ratio; see Kallenberg [2021] for conditional expectation, disintegration, and regular conditional distributions.

Subtracting one in (58) and using conditional Jensen's inequality gives the L^p -contraction explicitly:

$$\begin{aligned}\|F_h - 1\|_{L^p(\nu_h)}^p &= \mathbb{E} |\mathbb{E}[D_h - 1 \mid \sigma(X_0, X_h)]|^p \\ &\leq \mathbb{E} \{ \mathbb{E} [|D_h - 1|^p \mid \sigma(X_0, X_h)] \} \\ &= \|D_h - 1\|_{L^p(\mathbb{P})}^p.\end{aligned}$$

Hence

$$\|F_h - 1\|_{L^p(\nu_h)} \leq \|D_h - 1\|_{L^p(\mathbb{P})}. \quad (59)$$

We also need the conditional form of $\mu_h = F_h \nu_h$. Evaluating this measure identity on a rectangle $A \times B$ gives

$$\int_A Q_h(x, B) \pi(dx) = \int_A \int_B F_h(x, y) S_h(x, dy) \pi(dx). \quad (60)$$

Fix a countable π -system $\mathcal{C}$ that generates the Borel σ -field on $\mathbb{R}^d$, for example, the finite intersections of open balls with rational centers and radii, together with $\mathbb{R}^d$. For each $B \in \mathcal{C}$, equality (60) for every Borel A implies equality of the two integrands for π -almost every x , i.e.,

$$Q_h(x, B) = \int_B F_h(x, y) S_h(x, dy).$$

Since $\mathcal{C}$ is countable, there is one π -null set outside which this holds simultaneously for every $B \in \mathcal{C}$. For each fixed x outside that null set, both sides are measures in B , and the monotone-class theorem extends their equality from $\mathcal{C}$ to all Borel sets. Thus, for π -almost every x ,

$$Q_h(x, dy) = F_h(x, y) S_h(x, dy). \quad (61)$$

By reversibility, ν_h is symmetric. Hence the transpose measure $\mu_h^\top(dx, dy) = \mu_h(dy, dx)$ has density $F_h(y, x)$ with respect to ν_h . Since the target and proposal densities are strictly positive, the Radon–Nikodym chain rule gives

$$F_h(y, x) = \frac{e^{-U(y)} q_h(y, x)}{e^{-U(x)} q_h(x, y)} F_h(x, y) \quad \nu_h\text{-almost everywhere.}$$

The definition of the Metropolis–Hastings acceptance probability therefore implies

$$F_h(x, y) \alpha_h(x, y) = \min\{F_h(x, y), F_h(y, x)\} \quad \nu_h\text{-almost everywhere.}$$

This is the usual Metropolis–Hastings minimum identity [Tierney, 1998, Billera and Diaconis, 2001].

Because $\nu_h(dx, dy) = \pi(dx) S_h(x, dy)$, Tonelli's theorem shows that the preceding identity holds for $S_h(x, \cdot)$ -almost every y , for π -almost every x . Using (61), we obtain, for π -almost every x ,

$$\begin{aligned}r_h(x) &= 1 - \int \alpha_h(x, y) Q_h(x, dy) \\ &= 1 - \int \alpha_h(x, y) F_h(x, y) S_h(x, dy) \\ &= \int [1 - \min\{F_h(x, y), F_h(y, x)\}] S_h(x, dy) \\ &\leq \int (|F_h(x, y) - 1| + |F_h(y, x) - 1|) S_h(x, dy).\end{aligned}$$

Define

$$A_h(x) = \int |F_h(x, y) - 1| S_h(x, dy), \quad B_h(x) = \int |F_h(y, x) - 1| S_h(x, dy).$$

Then $r_h \leq A_h + B_h$. Jensen's inequality for the probability measure $S_h(x, dy)$ gives

$$\begin{aligned} \|A_h\|_{L^p(\pi)}^p &\leq \int |F_h(x, y) - 1|^p \nu_h(dx, dy), \\ \|B_h\|_{L^p(\pi)}^p &\leq \int |F_h(y, x) - 1|^p \nu_h(dx, dy). \end{aligned}$$

The first right-hand side is $\|F_h - 1\|_{L^p(\nu_h)}^p$; the second is the same because ν_h is symmetric. Minkowski's inequality therefore yields

$$\|r_h\|_{L^p(\pi)} \leq \|A_h\|_{L^p(\pi)} + \|B_h\|_{L^p(\pi)} \leq 2 \|F_h - 1\|_{L^p(\nu_h)}. \quad (62)$$

Combining (62), (59), and Lemma B.5 gives

$$\|r_h\|_{L^p(\pi)} \leq 2 \|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh \sqrt{p(d+p)},$$

after absorbing the factor two into the universal constant. This is (34). $\square$

B.4 Proving Proposition 3.2

Proof. We begin with the first of the two assertions. Let c be the step-size constant in Proposition B.1, and choose $0 < c_r \leq \min\{c, 1\}$. Fix $p \geq 1$ and t satisfying (11). Then Proposition B.1 applies to every $h \in [t/2, t]$, and

$$Lt \leq \frac{c_r}{\sqrt{p(d+p)}} \leq \frac{1}{\sqrt{2}} < 2.$$

Define the rejection probability averaged over $[t/2, t]$ by

$$\bar{r}_t(x) = \frac{2}{t} \int_{t/2}^t r_h(x) dh.$$

Minkowski's integral inequality and Proposition B.1 give

$$\begin{aligned} \|\bar{r}_t\|_{L^p(\pi)} &\leq \frac{2}{t} \int_{t/2}^t \|r_h\|_{L^p(\pi)} dh \\ &\leq CL \sqrt{p(d+p)} \frac{2}{t} \int_{t/2}^t h dh = \frac{3C}{4} Lt \sqrt{p(d+p)}. \end{aligned}$$

Set

$$G_{t,p} = \{x : \bar{r}_t(x) \leq 1/3\}.$$

Markov's inequality yields

$$\pi(G_{t,p}^c) \leq \left(\frac{9C}{4} Lt \sqrt{p(d+p)} \right)^p,$$

which is (12) after renaming the universal constant.

We next compare nearby proposal laws. Because U is convex and has L -Lipschitz gradient, the Baillon–Haddad cocoercivity inequality states that [Nesterov, 2004, Theorem 2.1.5]

$$\langle \nabla U(x) - \nabla U(y), x - y \rangle \geq \frac{1}{L} \|\nabla U(x) - \nabla U(y)\|^2.$$

See also [Bauschke and Combettes \[2010\]](#). Consequently, for $h \leq t$,

$$\begin{aligned} & \|x - h\nabla U(x) - y + h\nabla U(y)\|^2 \\ &= \|x - y\|^2 - 2h \langle x - y, \nabla U(x) - \nabla U(y) \rangle + h^2 \|\nabla U(x) - \nabla U(y)\|^2 \\ &\leq \|x - y\|^2. \end{aligned}$$

Note that we have used $h \leq t \leq 2/L$. Thus the proposal mean map $x \mapsto x - h\nabla U(x)$ is nonexpansive. Two Gaussian laws with covariance $2hI_d$ and means μ_x, μ_y have Kullback–Leibler divergence $\|\mu_x - \mu_y\|^2/(4h)$. Pinsker’s inequality [[Tsybakov, 2009](#), Lemma 2.5] therefore gives

$$\text{TV}(Q_h(x, \cdot), Q_h(y, \cdot)) \leq \frac{\|x - y\|}{\sqrt{8h}} \leq \frac{\|x - y\|}{2\sqrt{t}}, \quad h \in [t/2, t]. \quad (63)$$

For fixed x , write

$$R_{h,x}(\mathrm{d}y) = [1 - \alpha_h(x, y)] Q_h(x, \mathrm{d}y).$$

This measure has total mass $r_h(x)$, and, because $Q_h(x, \cdot)$ has a Lebesgue density, it gives no mass to $\{x\}$. Hence

$$P_h(x, \cdot) - Q_h(x, \cdot) = r_h(x)\delta_x - R_{h,x}$$

is the Jordan decomposition into mutually singular positive and negative parts. This proves

$$\text{TV}(P_h(x, \cdot), Q_h(x, \cdot)) = r_h(x). \quad (64)$$

Total variation is convex under mixing: for probability kernels M_h, N_h and a probability measure ρ ,

$$\text{TV}\left(\int M_h \mathrm{d}\rho, \int N_h \mathrm{d}\rho\right) \leq \int \text{TV}(M_h, N_h) \mathrm{d}\rho.$$

Applying this fact three times, together with the triangle inequality, (64), and (63), gives, for $x, y \in G_{t,p}$ with $\|x - y\| \leq \sqrt{t}/16$,

$$\begin{aligned} \text{TV}(\mathcal{K}_t(x, \cdot), \mathcal{K}_t(y, \cdot)) &\leq \bar{r}_t(x) + \frac{2}{t} \int_{t/2}^t \text{TV}(Q_h(x, \cdot), Q_h(y, \cdot)) \mathrm{d}h + \bar{r}_t(y) \\ &\leq \frac{1}{3} + \frac{1}{32} + \frac{1}{3} = \frac{67}{96} < \frac{3}{4}. \end{aligned}$$

This proves the first assertion of [Proposition 3.2](#). We note that this type of argument is common in the MCMC literature; see, e.g., the proof of Theorem 3 in [Wu et al. \[2022\]](#).

It remains to prove the final assertion. Let $Z \sim N_d(0, I_d)$, and $Y = x - h\nabla U(x) + \sqrt{2h}Z$. We first derive the log-ratio identity used below. Since, for $a, b \in \mathbb{R}^d$,

$$q_h(a, b) = (4\pi h)^{-d/2} \exp\left[-\frac{\|b - a + h\nabla U(a)\|^2}{4h}\right],$$

one has

$$\begin{aligned} \log \frac{e^{-U(Y)} q_h(Y, x)}{e^{-U(x)} q_h(x, Y)} &= U(x) - U(Y) \\ &\quad + \frac{1}{4h} \left(\|Y - x + h\nabla U(x)\|^2 - \|x - Y + h\nabla U(Y)\|^2 \right). \end{aligned}$$

Expanding the two squared norms and collecting the gradient and Gaussian terms gives

$$\begin{aligned} \log \frac{e^{-U(Y)} q_h(Y, x)}{e^{-U(x)} q_h(x, Y)} &= U(x) - U(Y) - \langle \nabla U(Y), x - Y \rangle - \frac{h}{4} \|\nabla U(Y) - \nabla U(x)\|^2 \\ &\quad - \sqrt{\frac{h}{2}} \langle Z, \nabla U(Y) - \nabla U(x) \rangle. \end{aligned} \quad (65)$$

The Baillon–Haddad inequality [Nesterov, 2004, Theorem 2.1.5] gives

$$U(x) - U(Y) - \langle \nabla U(Y), x - Y \rangle \geq \frac{1}{2L} \|\nabla U(Y) - \nabla U(x)\|^2.$$

If $h \leq 1/L$, the right-hand side of (65) is therefore at least

$$\frac{1}{4L} \|\nabla U(Y) - \nabla U(x)\|^2 - \sqrt{\frac{h}{2}} \|Z\| \|\nabla U(Y) - \nabla U(x)\| \geq -\frac{Lh}{2} \|Z\|^2.$$

It follows that

$$\alpha_h(x, Y) \geq e^{-Lh\|Z\|^2/2}, \quad \mathbb{E}[\alpha_h(x, Y)] \geq (1 + Lh)^{-d/2} \geq e^{-Lhd/2}.$$

Let $t' \in (0, 1/(2Ld)]$. Then every $h \in [t'/2, t']$ has rejection probability at most $1 - e^{-1/4}$, uniformly in x . Combining two such rejection bounds with (63), we obtain, whenever $\|x - y\| \leq \sqrt{t'}/16$,

$$\text{TV}(\mathcal{K}_{t'}(x, \cdot), \mathcal{K}_{t'}(y, \cdot)) \leq 2(1 - e^{-1/4}) + \frac{1}{32} < \frac{3}{4}.$$

This proves the final assertion of Proposition 3.2 and completes the proof. $\square$

C Proof of Proposition 3.3

We repeatedly use the following notation for the standard Gaussian distribution:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad \Phi(x) = \int_{-\infty}^x \varphi(v) \, dv, \quad \bar{\Phi}(x) = 1 - \Phi(x).$$

C.1 Some basic properties of Gaussian distribution functions

The Gaussian enlargement inequality (13), justified above under the first-order assumptions, states that

$$\pi(A^r) \geq \Phi(\Phi^{-1}(\pi(A)) + \sqrt{m}r).$$

We analyze the map $(q, s) \mapsto \Phi(\Phi^{-1}(q) + s)$.

We begin with a Mills-ratio bound.

Lemma C.1. *Let $a \geq 0$. Then*

$$\frac{\varphi(a)}{1+a} \leq \bar{\Phi}(a) \leq \frac{2\varphi(a)}{1+a}. \quad (66)$$

Moreover,

$$\log \frac{1}{\bar{\Phi}(a)} \leq (1+a)^2. \quad (67)$$

Proof. Let

$$g(a) = \bar{\Phi}(a) - \frac{\varphi(a)}{1+a}.$$

Then $g(a) \rightarrow 0$ as $a \rightarrow \infty$, and direct differentiation gives

$$g'(a) = -\frac{a\varphi(a)}{(1+a)^2} \leq 0.$$

Thus $g(a) \geq 0$, proving the lower bound in (66). Similarly, for

$$f(a) = \frac{2\varphi(a)}{1+a} - \bar{\Phi}(a),$$

we have $f(a) \rightarrow 0$ as $a \rightarrow \infty$ and

$$f'(a) = -\frac{(1+a^2)\varphi(a)}{(1+a)^2} < 0.$$

Hence $f(a) \geq 0$, proving the upper bound. Classical sharper Mills bounds for $a > 0$ go back at least to [Gordon \[1941\]](#).

The lower bound in (66) implies

$$\log \frac{1}{\bar{\Phi}(a)} \leq \frac{a^2}{2} + \log(1+a) + \frac{1}{2} \log(2\pi).$$

The difference between $(1+a)^2$ and the right-hand side is positive at $a = 0$, and its derivative is

$$2+a - \frac{1}{1+a} > 0.$$

This proves (67). □

Lemma C.2. *Let $q \in (0, 1/2]$ and $s \geq 0$. Then*

$$\Phi(\Phi^{-1}(q) + s) - q \geq \frac{q}{4} \min \left\{ 1, s\sqrt{\log(1/q)} \right\}.$$

Proof. Write $\Phi^{-1}(q) = -a$, so $a \geq 0$ and $q = \bar{\Phi}(a)$. Then

$$\Phi(-a+s) - \Phi(-a) = \int_0^s \varphi(a-v) dv.$$

For $0 \leq v \leq \min\{s, 1\}$,

$$\frac{\varphi(a-v)}{\varphi(a)} = e^{av-v^2/2} \geq e^{-1/2}.$$

By the upper Mills bound in (66), $\varphi(a) \geq q(1+a)/2$. Hence

$$\begin{aligned} \Phi(-a+s) - \Phi(-a) &\geq \frac{e^{-1/2}}{2} q(1+a) \min\{s, 1\} \\ &\geq \frac{e^{-1/2}}{2} q \min\{1, s(1+a)\} \\ &\geq \frac{q}{4} \min \left\{ 1, s\sqrt{\log(1/q)} \right\}. \end{aligned}$$

The second line uses $(1+a) \min\{s, 1\} \geq \min\{1, s(1+a)\}$; the third uses (67) and $e^{-1/2}/2 > 1/4$. □

C.2 Proving Proposition 3.3

Proof. If $r = 0$, the right-hand side of (14) is zero, so the conclusion is immediate. Assume $r > 0$ and, without loss of generality, that $\pi(A) = q$. Since $\text{dist}(A, B) \geq r$, the open enlargement A^r is disjoint from B , and hence $A^r \subseteq A \cup C$. Combining (13) and Lemma C.2 gives

$$\begin{aligned}\pi(C) &\geq \pi(A^r) - \pi(A) \\ &\geq \Phi(\Phi^{-1}(q) + \sqrt{m}r) - q \\ &\geq \frac{q}{4} \min \left\{ 1, r\sqrt{m \log(1/q)} \right\}.\end{aligned}$$

This is exactly (14). □

D Proof of Proposition 3.4

D.1 A generic conductance bound

We begin with a generic one-step flow bound for kernels that satisfy a close-coupling condition under the Gaussian isoperimetric inequality in Proposition 3.3. Combining close coupling with an isoperimetric inequality to lower-bound one-step flows, and hence conductance, is a standard technique. Related defective- or truncated-conductance arguments appear in Dwivedi et al. [2019], Wu et al. [2022], Chen and Gatmiry [2023], Liu and Tong [2026].

Lemma D.1. *Let K be π -reversible, let $t > 0$, and let $G \subseteq \mathbb{R}^d$ satisfy*

$$x, y \in G, \quad \|x - y\| \leq \frac{\sqrt{t}}{16} \implies \text{TV}(K(x, \cdot), K(y, \cdot)) \leq \frac{3}{4}.$$

Let S be a measurable set such that $0 < \pi(S) \leq 1/2$, and suppose that

$$\pi(G^c) \leq 2^{-12} \pi(S) \min \left\{ 1, \sqrt{mt \log[1/\pi(S)]} \right\}. \quad (68)$$

Then

$$\mathcal{J}_K(S, S^c) \geq 2^{-12} \pi(S) \min \left\{ 1, \sqrt{mt \log[1/\pi(S)]} \right\}. \quad (69)$$

Proof. Fix a measurable S with $0 < \pi(S) \leq 1/2$. Let

$$\beta = \min \left\{ 1, \sqrt{mt \log[1/\pi(S)]} \right\}.$$

Suppose, for contradiction, that

$$\mathcal{J}_K(S, S^c) < 2^{-12} \beta \pi(S). \quad (70)$$

Define

$$\begin{aligned}S_1 &= \{x \in S \cap G : K(x, S^c) < 1/8\}, \\ S_2 &= \{y \in S^c \cap G : K(y, S) < 1/8\}.\end{aligned}$$

For $x \in S_1$ and $y \in S_2$,

$$\text{TV}(K(x, \cdot), K(y, \cdot)) \geq K(x, S) - K(y, S) > \frac{3}{4}.$$

The close-coupling condition therefore implies

$$\text{dist}(S_1, S_2) \geq \frac{\sqrt{t}}{16}. \quad (71)$$

Markov's inequality and reversibility give

$$\pi(S_1^c \cap S \cap G) \leq 8\mathcal{J}_K(S, S^c), \quad \pi(S_2^c \cap S^c \cap G) \leq 8\mathcal{J}_K(S, S^c). \quad (72)$$

Thus,

$$\begin{aligned} \pi(S_1) &\geq \pi(S) - \pi(S \cap G^c) - 8\mathcal{J}_K(S, S^c), \\ \pi(S_2) &\geq 1 - \pi(S) - \pi(S^c \cap G^c) - 8\mathcal{J}_K(S, S^c). \end{aligned}$$

By (68) and the assumed upper bound (70),

$$\pi(S_1) > \pi(S) - 9 \cdot 2^{-12} \beta \pi(S), \quad \pi(S_2) > 1 - \pi(S) - 9 \cdot 2^{-12} \beta \pi(S).$$

Since $0 < \beta \leq 1$, $9 \cdot 2^{-12} < 1/2$, and $1 - \pi(S) \geq \pi(S)$, it follows that

$$\pi(S_1) \geq \frac{\pi(S)}{2}, \quad \pi(S_2) \geq \frac{\pi(S)}{2}. \quad (73)$$

Set $q' = \min\{\pi(S_1), \pi(S_2)\}$. Because $S_1 \subseteq S$,

$$\frac{\pi(S)}{2} \leq q' \leq \pi(S) \leq \frac{1}{2}.$$

Apply [Proposition 3.3](#) to S_1, S_2 , and their complement. Since $q' \leq \pi(S)$, one has $\log(1/q') \geq \log[1/\pi(S)]$. Using (71), (73), and $\min\{1, x/16\} \geq \min\{1, x\}/16$, we obtain

$$\begin{aligned} \pi((S_1 \cup S_2)^c) &\geq \frac{q'}{4} \min \left\{ 1, \frac{1}{16} \sqrt{mt \log(1/q')} \right\} \\ &\geq \frac{\pi(S)}{8} \cdot \frac{1}{16} \min \left\{ 1, \sqrt{mt \log[1/\pi(S)]} \right\} = \frac{\beta \pi(S)}{128}. \end{aligned} \quad (74)$$

On the other hand, by (72), (68), and (70),

$$\begin{aligned} \pi((S_1 \cup S_2)^c) &= \pi(G^c) + \pi(G \cap S \cap S_1^c) + \pi(G \cap S^c \cap S_2^c) \\ &\leq \pi(G^c) + 16\mathcal{J}_K(S, S^c) \\ &< 17 \cdot 2^{-12} \beta \pi(S) < \frac{\beta \pi(S)}{128}, \end{aligned}$$

contradicting (74). Therefore the assumed upper bound (70) is impossible, and (69) holds. $\square$

D.2 A technical lemma

The proof of [Proposition 3.4](#) via [Lemma D.1](#) requires one additional technical lemma.

Lemma D.2. *Let c_r, C_r be the constants in [Proposition 3.2](#). Fix a number $b_0 > 0$ satisfying*

$$b_0 \leq \min \left\{ \frac{1}{2}, c_r, \frac{1}{16C_r} \right\}. \quad (75)$$

Let $p \geq \tilde{p}_* := A_0 p_*$, $\theta \in (0, 1]$, and

$$t = \frac{\theta b_0}{L\sqrt{p(d+p)}},$$

where A_0 satisfies

$$A_0 \geq \frac{(49/4) \log 2 + (1/2) \log(2/b_0)}{\log 16 - (1/2)}. \quad (76)$$

Then, for every $u \in [\log 2, p/2]$,

$$\left(\frac{\theta}{16}\right)^p \leq 2^{-12} e^{-u} \sqrt{mtu}. \quad (77)$$

Moreover, $mtu \leq b_0/2 < 1$ throughout this range.

Proof. Since $b_0 \leq 1/2$, the bound (76) implies $A_0 \geq 2$. Consequently, since $p_* \geq 1$,

$$p \geq \tilde{p}_* \geq A_0 \geq 2 > 2 \log 2 > 1,$$

so $[\log 2, p/2]$ is nonempty. The last assertion follows directly from $m/L = 1/\kappa$:

$$mtu = \frac{\theta b_0 u}{\kappa \sqrt{p(d+p)}} \leq \frac{\theta b_0}{2\kappa} \sqrt{\frac{p}{d+p}} \leq \frac{b_0}{2}.$$

We focus on proving (77). Since $u \mapsto e^{-u} \sqrt{u}$ decreases on $[1/2, \infty)$, the right-hand side of (77) is smallest at $u = p/2$. At $u = p/2$ it equals

$$2^{-12} e^{-p/2} \left(\frac{\theta b_0}{2\kappa}\right)^{1/2} \left(\frac{p}{d+p}\right)^{1/4}.$$

Dividing the desired inequality at $u = p/2$ by $\theta^{1/2}$ shows that its left-hand side contains the factor $\theta^{p-1/2}$. Since $p \geq 1$ and $0 < \theta \leq 1$, this factor is at most one. It is therefore sufficient that

$$p \left(\log 16 - \frac{1}{2} \right) \geq 12 \log 2 + \frac{1}{2} \log \frac{2\kappa}{b_0} + \frac{1}{4} \log \frac{d+p}{p}. \quad (78)$$

Since $p \geq 1$ and $d \geq 1$,

$$\log \frac{d+p}{p} \leq \log(2d) = \log d + \log 2.$$

The right-hand side of (78) is therefore at most

$$\frac{49}{4} \log 2 + \frac{1}{2} \log \frac{2}{b_0} + \frac{1}{2} \log \kappa + \frac{1}{4} \log d.$$

By (76) and $b_0 \leq 1/2$,

$$A_0 \left(\log 16 - \frac{1}{2} \right) \geq \frac{49}{4} \log 2 + \frac{1}{2} \log \frac{2}{b_0} \geq \frac{53}{4} \log 2 > \frac{1}{2}.$$

Since $\log d$ and $\log \kappa$ are nonnegative, these bounds and $p \geq \tilde{p}_*$ give

$$\begin{aligned} p \left(\log 16 - \frac{1}{2} \right) &\geq A_0 \left(\log 16 - \frac{1}{2} \right) (1 + \log d + \log \kappa) \\ &\geq \frac{49}{4} \log 2 + \frac{1}{2} \log \frac{2}{b_0} + \frac{1}{4} \log d + \frac{1}{2} \log \kappa. \end{aligned}$$

This proves (78), uniformly in all parameters. $\square$

D.3 Proving Proposition 3.4

Proof. We begin with the first assertion. Let A_0 and b_0 be universal constants satisfying (75) and (76). Let $p \geq \tilde{p}_\star := A_0 p_\star$, $\theta \in (0, 1]$, and

$$t = \frac{\theta b_0}{L\sqrt{p(d+p)}}.$$

Let $S \subseteq \mathbb{R}^d$ be measurable with $e^{-p/2} \leq \pi(S) \leq 1/2$. By Lemma D.2, since $\log[1/\pi(S)] \in [\log 2, p/2]$,

$$\left(\frac{\theta}{16}\right)^p \leq 2^{-12}\pi(S)\sqrt{mt \log[1/\pi(S)]} = 2^{-12}\pi(S) \min \left\{1, \sqrt{mt \log[1/\pi(S)]}\right\}.$$

The equality follows from the final assertion of Lemma D.2, namely, $mt \log[1/\pi(S)] \leq 1$. Because $b_0 \leq c_r$ and $b_0 \leq 1/(16C_r)$, the first assertion of Proposition 3.2 yields a measurable set $G_{t,p} \subseteq \mathbb{R}^d$ such that

$$\pi(G_{t,p}^c) \leq \left[C_r L \frac{\theta b_0}{L\sqrt{p(d+p)}} \sqrt{p(d+p)} \right]^p \leq \left(\frac{\theta}{16}\right)^p,$$

and

$$x, y \in G_{t,p}, \quad \|x - y\| \leq \frac{\sqrt{t}}{16} \implies \text{TV}(\mathcal{K}_t(x, \cdot), \mathcal{K}_t(y, \cdot)) \leq \frac{3}{4}.$$

Then, by Lemma D.1,

$$\mathcal{J}_{\mathcal{K}_t}(S, S^c) \geq 2^{-12}\pi(S) \min \left\{1, \sqrt{mt \log[1/\pi(S)]}\right\} = 2^{-12}\pi(S)\sqrt{mt \log[1/\pi(S)]}.$$

This proves the first assertion.

For the second assertion, let $t' \in (0, 1/(2Ld)]$ and let $S \subseteq \mathbb{R}^d$ be measurable with $0 < \pi(S) \leq 1/2$. By the second assertion of Proposition 3.2,

$$x, y \in \mathbb{R}^d, \quad \|x - y\| \leq \frac{\sqrt{t'}}{16} \implies \text{TV}(\mathcal{K}_{t'}(x, \cdot), \mathcal{K}_{t'}(y, \cdot)) \leq \frac{3}{4}.$$

Applying Lemma D.1 with $G = \mathbb{R}^d$ yields

$$\mathcal{J}_{\mathcal{K}_{t'}}(S, S^c) \geq 2^{-12}\pi(S) \min \left\{1, \sqrt{mt' \log[1/\pi(S)]}\right\}.$$

This proves the second assertion. □

E Proof of Lemma 3.5

The result can also be derived from the general Cheeger inequality of Chen and Wang [2000]. The proof below is direct and self-contained; we include the details because the precise dependence on β_j and γ_j is needed later.

Proof. For $j = 1, \dots, N$, define the symmetric measure

$$J_j(dx, dy) = \varpi(dx)K_j(x, dy).$$

The measure J_j is symmetric because K_j is ϖ -reversible. Both of its marginals equal ϖ .

We first prove

$$\sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g, g) \geq \frac{\|g\|_{L^2(\varpi)}^2}{2 \sum_{j=1}^N \beta_j^2 / \gamma_j} \quad (79)$$

for a nonnegative function $g \in L^2(\varpi)$ such that $\varpi(g > 0) \leq 1/2$. For $t \geq 0$, let

$$A_t = \{x : g(x)^2 > t\}.$$

Then $\varpi(A_t) \leq 1/2$, so the assumed fractional boundary inequality gives

$$\varpi(A_t) \leq \sum_{j=1}^N \beta_j \mathcal{J}_{K_j}(A_t, A_t^c).$$

Integrating this inequality over $t \geq 0$ and using the layer-cake identity yields

$$\begin{aligned} \|g\|_{L^2(\varpi)}^2 &= \int_0^\infty \varpi(A_t) dt \\ &\leq \sum_{j=1}^N \beta_j \int_0^\infty \mathcal{J}_{K_j}(A_t, A_t^c) dt. \end{aligned}$$

For each j , symmetry of J_j gives the coarea identity

$$\int_0^\infty \mathcal{J}_{K_j}(A_t, A_t^c) dt = \frac{1}{2} \iint |g(x)^2 - g(y)^2| J_j(dx, dy).$$

Indeed, for fixed x, y ,

$$\int_0^\infty \mathbf{1}_{\{g(x)^2 > t \geq g(y)^2\}} dt = [g(x)^2 - g(y)^2]_+,$$

and symmetry changes the integral of the positive part into one half of the integral of the absolute difference. Consequently,

$$\|g\|_{L^2(\varpi)}^2 \leq \frac{1}{2} \sum_{j=1}^N \beta_j \iint |g(x) - g(y)| |g(x) + g(y)| J_j(dx, dy).$$

Apply Cauchy–Schwarz simultaneously over the component index j and the pair (x, y) :

$$\begin{aligned} \|g\|_{L^2(\varpi)}^2 &\leq \frac{1}{2} \left\{ \sum_{j=1}^N \gamma_j \iint [g(x) - g(y)]^2 J_j(dx, dy) \right\}^{1/2} \\ &\quad \times \left\{ \sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} \iint [g(x) + g(y)]^2 J_j(dx, dy) \right\}^{1/2}. \end{aligned}$$

The first square root equals

$$\left[2 \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g, g) \right]^{1/2}.$$

For the second square root, the inequality

$$[g(x) + g(y)]^2 \leq 2g(x)^2 + 2g(y)^2$$

and the fact that both marginals of J_j equal ϖ give

$$\iint [g(x) + g(y)]^2 J_j(dx, dy) \leq 4 \|g\|_{L^2(\varpi)}^2.$$

It follows that

$$\|g\|_{L^2(\varpi)}^2 \leq \left[2 \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g, g) \right]^{1/2} \|g\|_{L^2(\varpi)} \left(\sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} \right)^{1/2}.$$

If $g = 0$ in $L^2(\varpi)$, there is nothing to prove. Otherwise, cancelling one factor of $\|g\|_{L^2(\varpi)}$ and squaring gives (79).

Now let $f \in L^2(\varpi)$, and let c be a median of f ; thus

$$\varpi(f \geq c) \geq \frac{1}{2}, \quad \varpi(f \leq c) \geq \frac{1}{2}.$$

Define the positive and negative parts of $f - c$:

$$g_+ = (f - c)_+, \quad g_- = (f - c)_-.$$

Then

$$\varpi(g_+ > 0) \leq \frac{1}{2}, \quad \varpi(g_- > 0) \leq \frac{1}{2}.$$

Moreover, for every x, y ,

$$[f(x) - f(y)]^2 \geq [g_+(x) - g_+(y)]^2 + [g_-(x) - g_-(y)]^2.$$

To see this, write

$$f - c = g_+ - g_-.$$

The two differences

$$g_+(x) - g_+(y) \quad \text{and} \quad g_-(x) - g_-(y)$$

have opposite signs, so the cross term in the square of their difference is nonnegative.

Therefore, by (79),

$$\begin{aligned} \mathcal{E}_P(f, f) &\geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(f, f) \\ &\geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g_+, g_+) + \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g_-, g_-) \\ &\geq \frac{\|g_+\|_{L^2(\varpi)}^2 + \|g_-\|_{L^2(\varpi)}^2}{2 \sum_{j=1}^N \beta_j^2 / \gamma_j}. \end{aligned}$$

Since g_+ and g_- have disjoint supports and $f - c = g_+ - g_-$,

$$\|g_+\|_{L^2(\varpi)}^2 + \|g_-\|_{L^2(\varpi)}^2 = \|f - c\|_{L^2(\varpi)}^2.$$

The mean minimizes $a \mapsto \|f - a\|_{L^2(\varpi)}^2$, so

$$\|f - c\|_{L^2(\varpi)}^2 \geq \text{Var}_{\varpi}(f).$$

We conclude that

$$\mathcal{E}_P(f, f) \geq \frac{\text{Var}_{\varpi}(f)}{2 \sum_{j=1}^N \beta_j^2 / \gamma_j}.$$

Taking the infimum over all nonconstant $f \in L^2(\varpi)$ proves the claim. $\square$

F Multiscale arithmetic

This appendix verifies a summation estimate used in [Section 4](#). The constants $b_0 \leq 1/2$ and $A_0 \geq 1$ are the universal choices made at the beginning of the proof of [Theorem 2.1](#) in [Section 4](#), and $\tilde{p}_\star = A_0 p_\star$.

Lemma F.1. *Assume $\tilde{p}_\star < d$, and define p_j, J, t_j, θ by (15)–(17). With*

$$\gamma_j = \frac{t_j}{2H}, \quad \phi_0^2 = 2^{-24} m t_0 \log 2, \quad \phi_j^2 = 2^{-27} m t_j p_j \quad (j \geq 1),$$

there is a universal $C \in (0, \infty)$ such that

$$\sum_{j=0}^J \frac{1}{\gamma_j \phi_j^2} \leq C \frac{H L^2 \tilde{p}_\star (d + \tilde{p}_\star)}{m \theta^2 b_0^2}.$$

Proof. For the first component,

$$\frac{1}{\gamma_0 \phi_0^2} = \frac{2^{25} H}{m t_0^2 \log 2} = \frac{2^{25} H L^2 \tilde{p}_\star (d + \tilde{p}_\star)}{m \theta^2 b_0^2 \log 2}.$$

For $j \geq 1$,

$$\frac{1}{\gamma_j \phi_j^2} = \frac{2^{28} H}{m t_j^2 p_j} = \frac{2^{28} H L^2 (d + p_j)}{m \theta^2 b_0^2}.$$

It remains to show

$$\tilde{p}_\star (d + \tilde{p}_\star) + \sum_{j=1}^J (d + p_j) \leq C \tilde{p}_\star (d + \tilde{p}_\star)$$

for a universal C . Because $\tilde{p}_\star < d$, one has $J \geq 1$. Minimality of J gives $4^{J-1} \tilde{p}_\star < d$, and hence

$$J < 1 + \log_4 \frac{d}{\tilde{p}_\star} \leq 1 + \log_4 d.$$

The geometric-series formula and (16) give

$$\sum_{j=1}^J p_j = \tilde{p}_\star \sum_{j=1}^J 4^j = \frac{4p_J - 4\tilde{p}_\star}{3} < \frac{16d}{3}.$$

Moreover, $\tilde{p}_\star \geq A_0(1 + \log d)$, $A_0 \geq 1$, and $\tilde{p}_\star \geq 1$ imply

$$1 + \log_4 d \leq 1 + \frac{\tilde{p}_\star}{A_0 \log 4} \leq \left(1 + \frac{1}{\log 4}\right) \tilde{p}_\star.$$

Therefore

$$\sum_{j=1}^J (d + p_j) = Jd + \sum_{j=1}^J p_j \leq C \tilde{p}_\star d$$

for some universal constant $C \in (0, \infty)$. The desired inequality then follows by enlarging C . $\square$

G A spectral gap for small fixed step sizes

The uniform acceptance estimate used in the proof of [Proposition 3.2](#) also yields a spectral-gap lower bound for fixed-step MALA. We record this consequence below.

Proposition G.1. *Suppose that U satisfies (3), and let $\kappa = L/m$. There is a universal constant $c \in (0, \infty)$ such that, for every*

$$0 < h \leq \frac{1}{2Ld},$$

the fixed-step MALA kernel satisfies

$$\text{Gap}_R(P_h) \geq cmh.$$

In particular, choosing $h = 1/(2Ld)$ gives

$$\text{Gap}_R(P_h) \geq \frac{c}{\kappa d}.$$

Proof. Fix $h \in (0, 1/(2Ld)]$. For $x \in \mathbb{R}^d$, let

$$Y = x - h\nabla U(x) + \sqrt{2h} Z, \quad Z \sim N_d(0, I_d).$$

Since $h \leq 1/L$, the argument following (65) in the proof of [Proposition 3.2](#) gives

$$1 - r_h(x) = \mathbb{E}[\alpha_h(x, Y)] \geq e^{-Lhd/2} \geq e^{-1/4}.$$

Thus $r_h(x) \leq 1 - e^{-1/4}$ for every $x \in \mathbb{R}^d$.

Because U is convex and has an L -Lipschitz gradient, the proposal mean map $x \mapsto x - h\nabla U(x)$ is nonexpansive whenever $h \leq 2/L$, as shown in the proof of [Proposition 3.2](#). The Gaussian comparison leading to (63) therefore gives

$$\text{TV}(Q_h(x, \cdot), Q_h(y, \cdot)) \leq \frac{\|x - y\|}{\sqrt{8h}}.$$

Combining this inequality with (64) and the triangle inequality, we find that, whenever $\|x - y\| \leq \sqrt{h}/16$,

$$\begin{aligned} \text{TV}(P_h(x, \cdot), P_h(y, \cdot)) &\leq r_h(x) + \text{TV}(Q_h(x, \cdot), Q_h(y, \cdot)) + r_h(y) \\ &\leq 2(1 - e^{-1/4}) + \frac{1}{16\sqrt{8}} < \frac{3}{4}. \end{aligned}$$

This overlap bound holds for all $x, y \in \mathbb{R}^d$.

Apply [Lemma D.1](#) with $K = P_h$, $t = h$, and $G = \mathbb{R}^d$. Its exceptional-set condition holds because $\pi(G^c) = 0$. Consequently, for every measurable S such that $0 < \pi(S) \leq 1/2$,

$$\mathcal{J}_{P_h}(S, S^c) \geq 2^{-12} \pi(S) \min \left\{ 1, \sqrt{mh \log[1/\pi(S)]} \right\}.$$

Since

$$\log[1/\pi(S)] \geq \log 2, \quad mh \log 2 \leq \frac{\log 2}{2\kappa d} < 1,$$

it follows that

$$\mathcal{J}_{P_h}(S, S^c) \geq 2^{-12} \sqrt{mh \log 2} \pi(S).$$

The one-kernel case of [Lemma 3.5](#), which is the usual Cheeger inequality [[Lawler and Sokal, 1988](#)], now gives

$$\text{Gap}_R(P_h) \geq \frac{1}{2} \left(2^{-12} \sqrt{mh \log 2} \right)^2 = 2^{-25} (\log 2) mh.$$

More explicitly, the lemma applies with $P = K_1 = P_h$, $\varpi = \pi$, $\gamma_1 = 1$, and $\beta_1 = 2^{12}/\sqrt{mh \log 2}$.

For $h = 1/(2Ld)$, the preceding bound becomes

$$\text{Gap}_R(P_h) \geq \frac{2^{-26} \log 2}{\kappa d}.$$

Both assertions therefore hold with $c = 2^{-26} \log 2$. □

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